

Image analysis: MicrolCS hands-on session

Deep dive into the statistical approaches used to validate potential standardised laboratory methods

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Biofilm
Regulatory
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Jožef Stefan
Institute
Ljubljana, Slovenia



Introduction



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Training school

The purple path is the worked *Listeria* example we run together.

Think about how it applies to your own work

The green path is your own work, carried from the start so that the two meet at the end.



DECIDE & FRAME · Obj 1–2

What to measure?

*defining your own question
defining your own label*



DESIGN & ACQUIRE · Obj 3

How to measure?

*planning your experiments
and imaging settings*



RUN MicroICS · Obj 4

How to analyse?

*independently running the
software*



READ & INTERPRET · Obj 5–6

How to interpret results?

*interpreting your own output
how can the output help me
to understand my dataset*



KNOW THE LIMITS · Obj 7

What are the limitations of the pipeline?

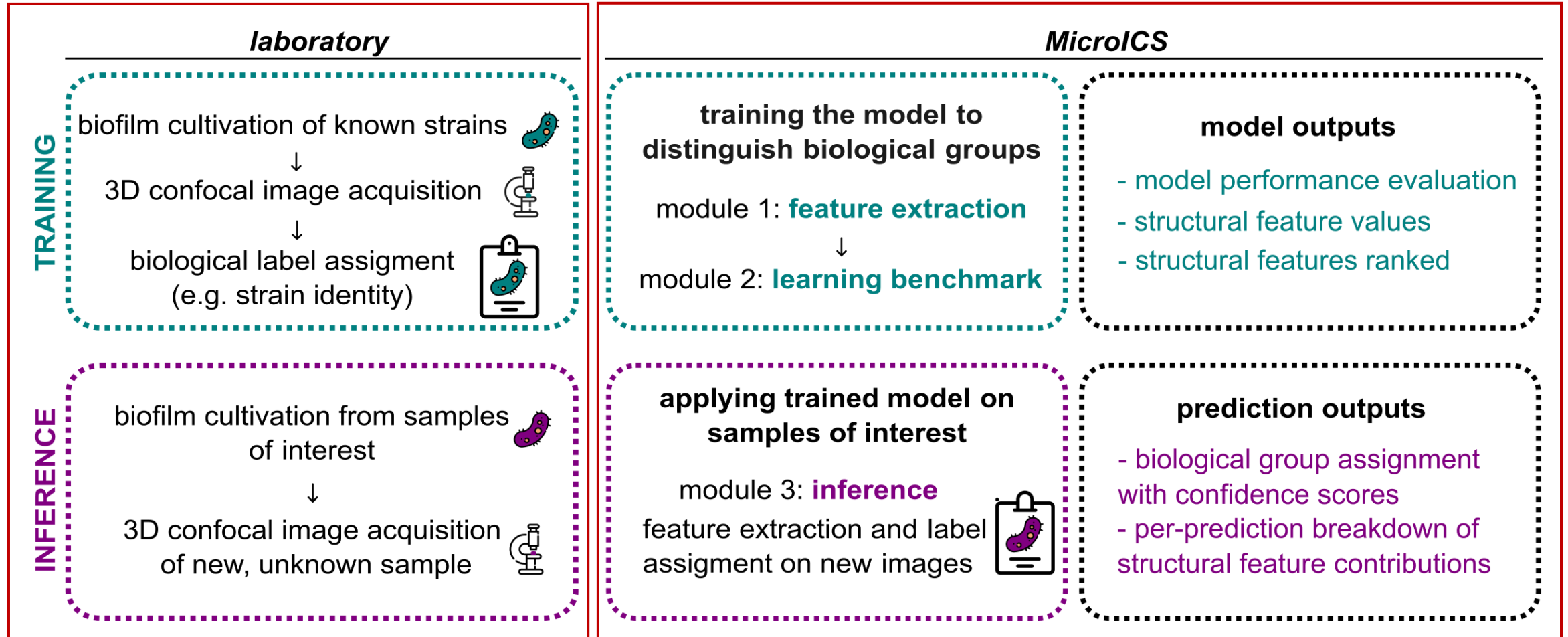
*identifying the follow-up
experiments*

The destination

You can see how to apply MicroICS to your own organism, treatment, and biological question to help you generate hypotheses for further work.

Here we do this in theory.

Here we try it during the session.



Use your cards to help us navigate:

- I'm fine, ready 
- please help, I'm stuck 
- please help, I'm lost in the space. 



Decide and frame

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Let's say...

...I have a question about something I've observed.

Can I use machine learning to answer it?

It depends — but if I've measured the right information, **almost certainly yes**.

But, there's a **catch**.

The catch with machine learning

If your data relates to your question, there's almost certainly a method that can help you find answers.

The tricky part is knowing which method fits your data. Just in this building there are a hundred ML researchers, each specializing in different techniques for different problems (questions).

In short: **there is no one-size-fits-all machine learning solution.**

Getting acquainted with machine learning

- Determine whether a **day is appropriate for a picnic**
 - Data-driver approach => Based on prior observations
- We are predicting a binary value, our **classes/labels** are:
YES and **NO**
- Observe some days and label them either **YES** or **NO**
- Based on these measurements what kind of model can we construct?
- **Majority class:**
 - Most days aren't picnic days, thus we always predict **NO**
 - In general, we are correct more often if we predict the most common class over any other

Class (picnic)	Prediction (picnic)
YES	NO
NO	NO
YES	NO
NO	NO
NO	NO
YES	NO
NO	NO
YES	NO
NO	NO
NO	NO
NO	NO
NO	NO

Mistakes are unavoidable

- Evaluating **how the model makes mistakes** offers key insights
- Common tool: **confusion matrix**
- Common metric: **accuracy**
 - The ratio of the number of correctly classified examples and the number of all examples
 - For majority class: **accuracy = 66.7%**

	Predicted NO	Predicted YES
Actual NO	8 (66.7%)	0 (0%)
Actual YES	4 (33.3%)	0 (0%)

Measuring the world

- Measure properties of the observed days (features)
- **Based on these properties**, can we construct a model?
- A simple rule:

IF Weather == "Rainy" ***THEN*** predict **NO**
ELSE predict **YES**

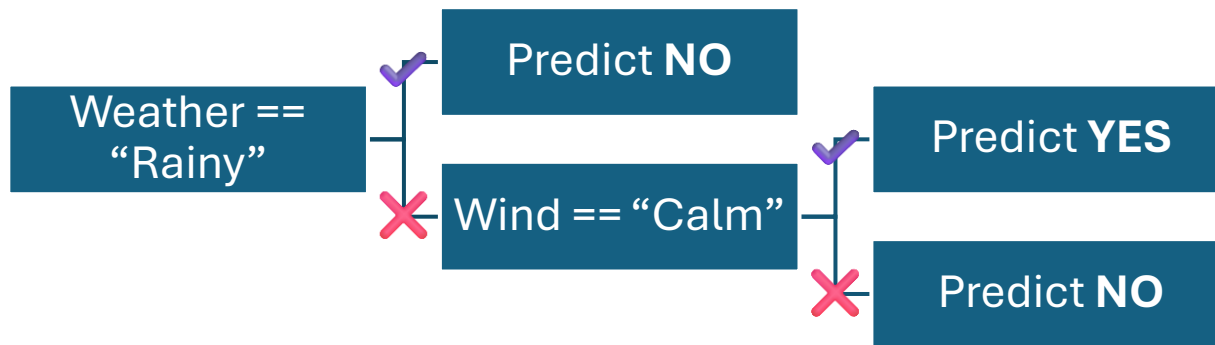
- Model evaluation:
 - accuracy = 83.3%

	Predicted NO	Predicted YES
Actual NO	6 (50.0%)	2 (16.7%)
Actual YES	0 (0%)	4 (33.3%)

Prediction (picnic)	Class (picnic)	Weather	Day	Holiday
YES	YES	Sunny	Monday	No
NO	NO	Rainy	Tuesday	No
YES	YES	Sunny	Wednesday	Yes
NO	NO	Rainy	Thursday	No
YES	NO	Sunny	Friday	No
YES	YES	Sunny	Saturday	No
NO	NO	Rainy	Sunday	No
YES	YES	Sunny	Monday	No
NO	NO	Rainy	Tuesday	No
YES	NO	Sunny	Wednesday	No
NO	NO	Rainy	Thursday	No
NO	NO	Rainy	Friday	No

Measuring more sets us up for success

- Measuring more **improves our odds** at getting a better model
 - More properties
 - More data points (examples)

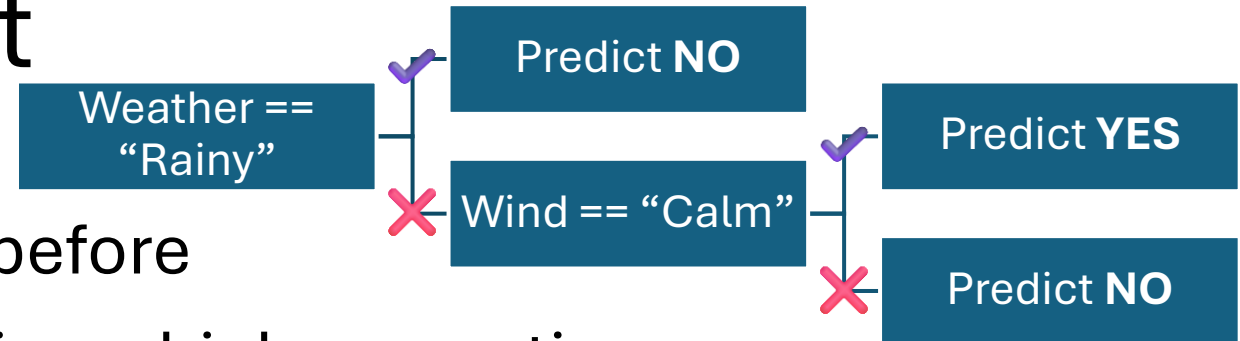


- Decision tree

	Predicted NO	Predicted YES
Actual NO	11 (68.8%)	0 (0%)
Actual YES	0 (0%)	5 (31.1%)

Prediction (picnic)	Class (picnic)	Weather	Day	Holiday	Wind	Sunrise
YES	YES	Sunny	Monday	No	Calm	6:26:12 am
NO	NO	Rainy	Tuesday	No	Stormy	6:27:27 am
YES	YES	Sunny	Wednesday	Yes	Calm	6:28:42 am
NO	NO	Rainy	Thursday	No	Breezy	6:29:57 am
NO	NO	Sunny	Friday	No	Stormy	6:31:12 am
YES	YES	Sunny	Saturday	No	Calm	6:32:27 am
NO	NO	Rainy	Sunday	No	Stormy	6:33:42 am
YES	YES	Sunny	Monday	No	Calm	6:34:57 am
NO	NO	Rainy	Tuesday	No	Breezy	6:36:11 am
NO	NO	Sunny	Wednesday	No	Stormy	6:37:26 am
NO	NO	Rainy	Thursday	No	Stormy	6:38:41 am
NO	NO	Rainy	Friday	No	Breezy	6:39:57 am
NO	NO	Rainy	Saturday	No	Stormy	6:41:12 am
YES	YES	Sunny	Sunday	No	Calm	6:42:27 am
NO	NO	Rainy	Monday	No	Breezy	6:43:42 am
NO	NO	Rainy	Tuesday	Yes	Stormy	6:44:58 am

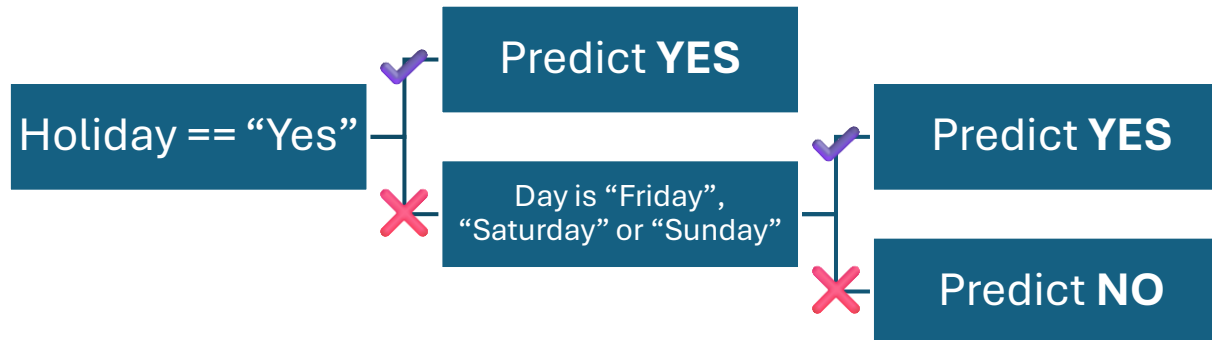
Changing our viewpoint



- Observe the decision tree from before
- **From the model alone**, determine which properties (features) are important for prediction?
- From model structure: **Weather** and **Wind** are important, but which is more important?
 - Weather selects **over all examples**
 - Wind selects over examples **where Weather == "Sunny"**
- Quantifying this gives us a **feature ranking**

Asking a different question

- Predict which days are **party days**
- **Reusing previous data...**
... but **changing the class**
and the **ground truth with it.**
- Leads to a **different model** and **different features** (Day, Holiday)



Class (party)	Weather	Day	Holiday	Wind	Sunrise
NO	Sunny	Monday	No	Calm	6:26:12 am
NO	Rainy	Tuesday	No	Stormy	6:27:27 am
YES	Sunny	Wednesday	Yes	Calm	6:28:42 am
NO	Rainy	Thursday	No	Breezy	6:29:57 am
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YES	Sunny	Sunday	No	Calm	6:42:27 am
NO	Rainy	Monday	No	Breezy	6:43:42 am
YES	Rainy	Tuesday	Yes	Stormy	6:44:58 am

Key takeaway:

Which question we choose impacts the model and which features are important.

Enter MicrolCS



We built MicrolCS to help you explore biofilm images.

MicrolCS takes your **images and your labels** (your question) and uses the images to **construct structural features*** and learns a **model for your labels**.

*In our toy example, we had to somehow decide which properties we measure — MicrolCS automatically learns and selects structural features from biofilm images.

Investigating **the model** reveals how well structural features separate your labels. **Feature ranking** tells you which features carry the signal of your labels. **Inference** tells you how new images fit into your labels.

Harnessing biofilm images

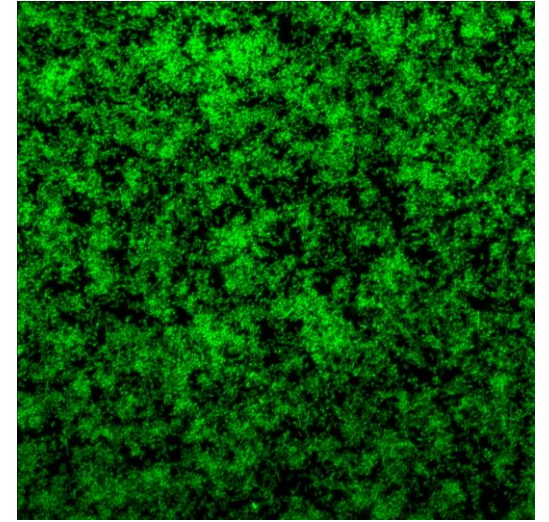
A 3D biofilm image contains **far more information than we can easily process by eye.**

We can see that two biofilms look different, but describing **how they differ — consistently and reproducibly** — is much harder.

MicroICS extracts and quantifies thousands of structural properties from each 3D image and then learns which combinations of properties help distinguish your labels.

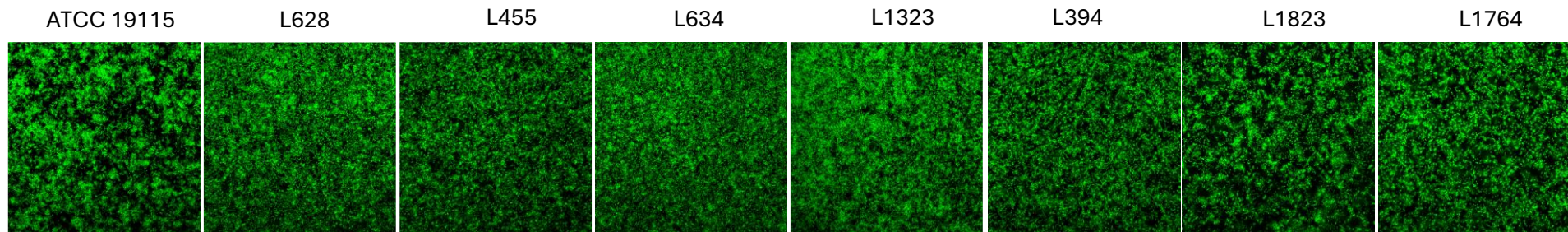
You don't need to decide in advance which features matter — the model finds the patterns from your labels.

The result is a **quantified, testable description** of what differs between the groups.

[illegible]

Asking the right questions with MicroICS

- **What kind of labels can we use for classifying biofilms?**
 - A **property that we want to study** — strain identity, treatment (different antibiotics), or source (isolation origin)
 - Using MicroICS, we always work under the assumption that **it can be differentiated from structural features**
- As we have seen before, we must **know the ground truth** (which images have which properties/labels) to **discover the underlying structural differences**



Leveraging ***your*** expertise with MicroICS

- The model tells you **which structural features carry the signal**, not what they mean biologically ...
... that **interpretation remains yours** and is where the **biology begins**.
- If you are **unsure whether your question is reflected in structure**, this is itself worth testing:
 - Given representative data, a **model with failing accuracy** is telling you something — either the **signal is not in the structure**, or **your dataset is not yet able to show it**.

Define your question:

Which property is dependent on biofilm structure?

Define your labels:

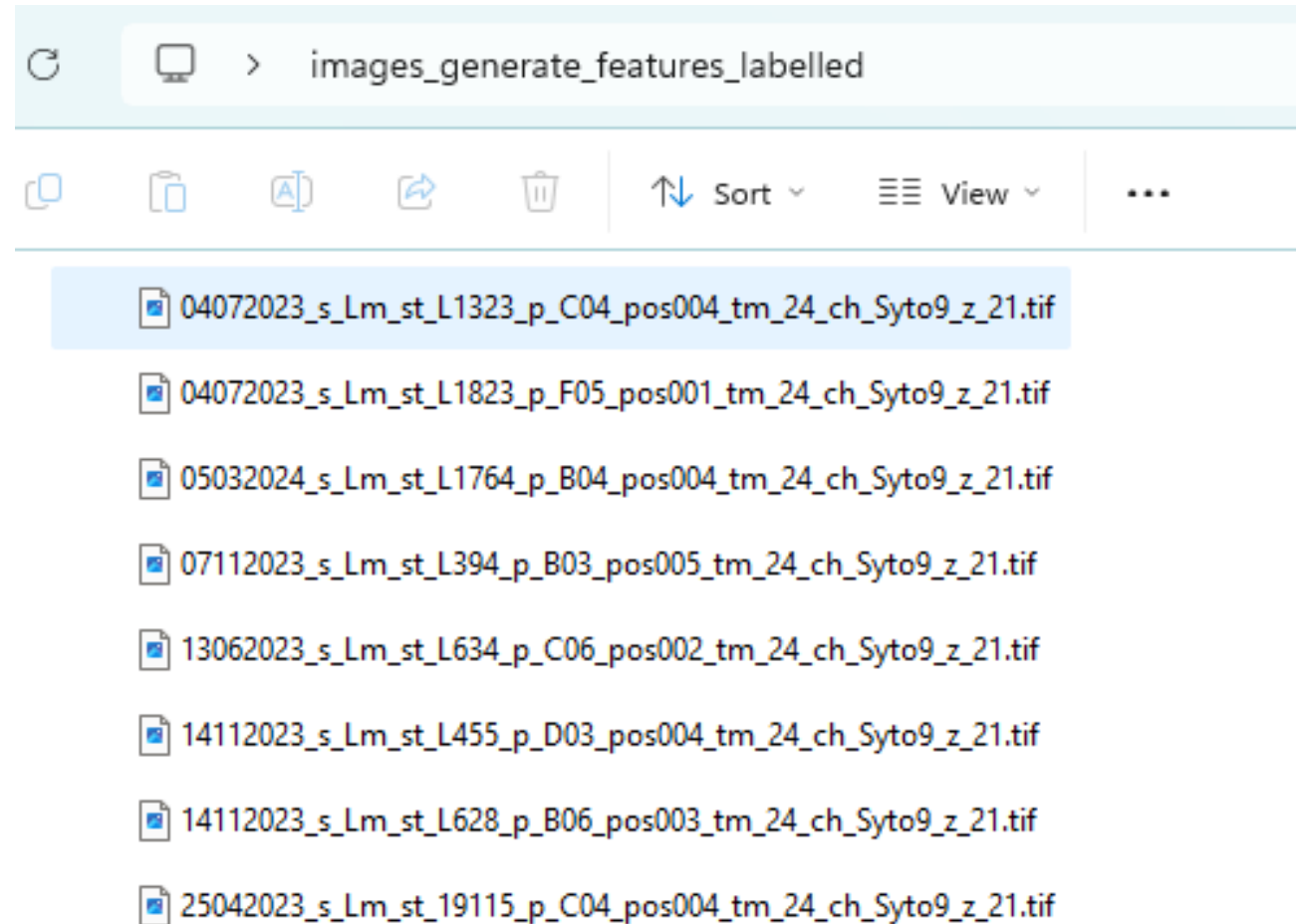
Which possible labels (values of your property) are you measuring?

Let's explore our example dataset together. What do you notice?

Task 1: Our goal is to train a model to distinguish between *Listeria* strains.

We prepared 1000 images of 8 strains (125 images per strain), and we named the images so that the strain name (label) is evident from the name.

Do you find it? Hint: the label is given in the name of each image after `_st_`. How many classes do we have?

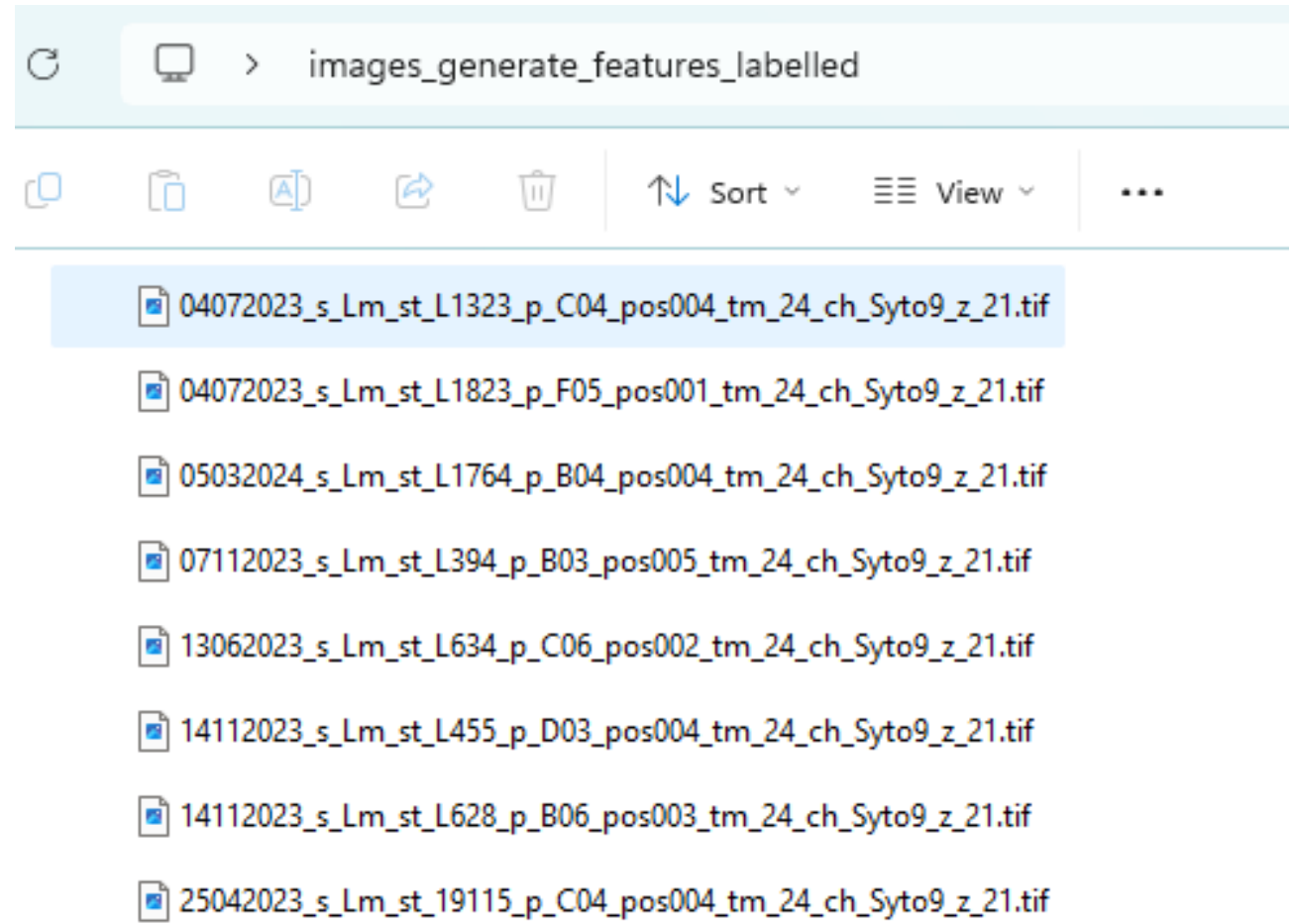


Let's explore our example dataset together. What do you notice?

Task 2: We want to use the same data set to train a model to distinguish between strains that are clinically relevant and clinically non-relevant.

Hint: clinically relevant strains are L628, L455, L634, L1823, ATCC 19115, and clinically non-relevant strains are L1764, L394, L1323.

- How many classes do we have in this case?
- How do we change the labels of images to train the model to distinguish between clinically relevant and non-relevant strains?



Think about your own research topic.

Is there a question where you suspect that biofilm structure plays a role, but nobody has looked at this yet, or it is hard to infer from the images you already have?



Design and acquire images

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Center for Advanced Optical microscopies (cNOM)



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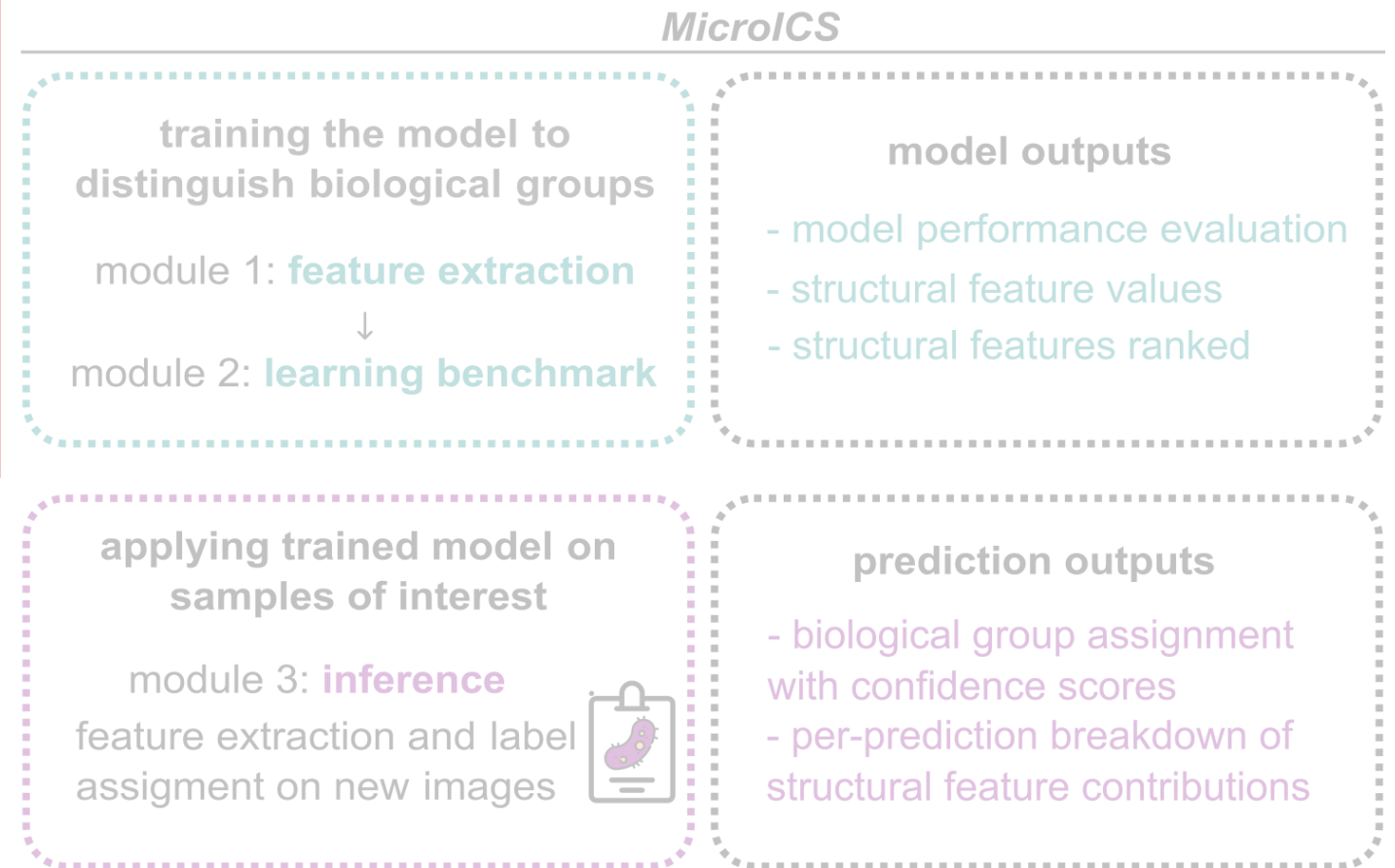
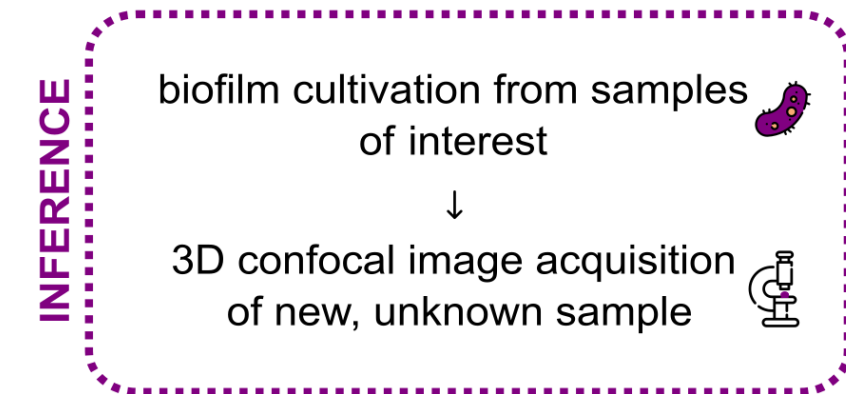
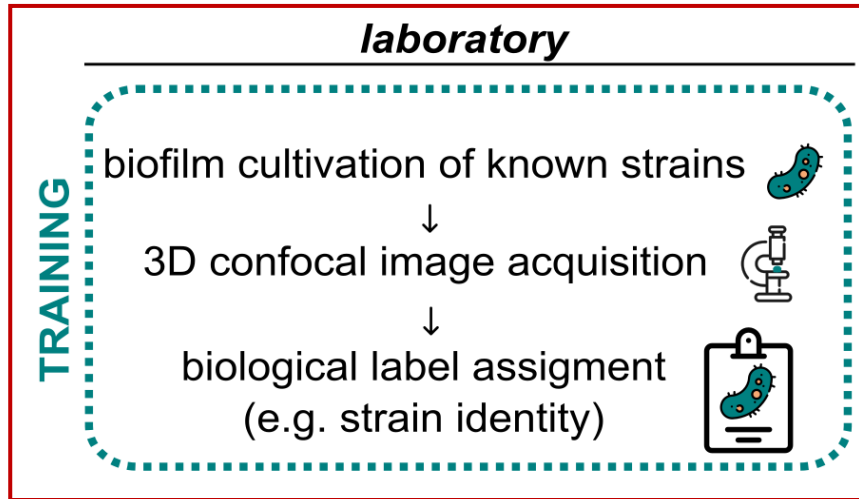


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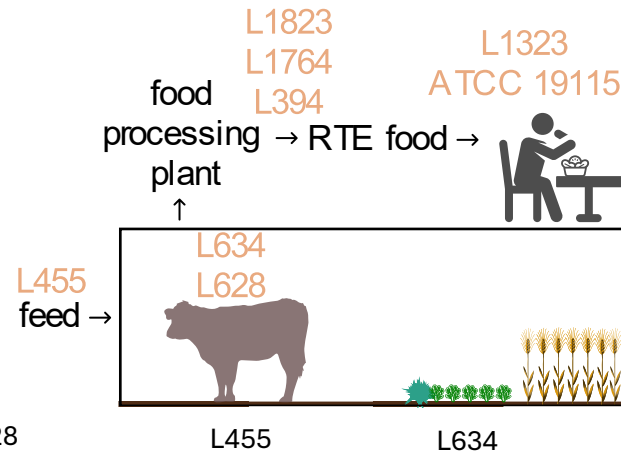
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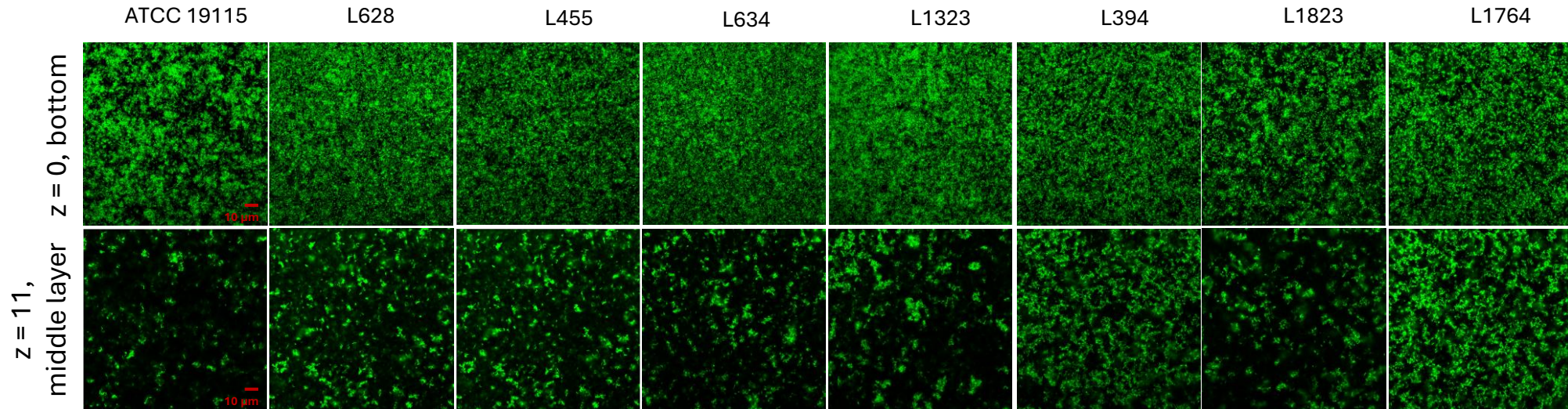


Where the data begin: **strains**, biofilms, and staining – then imaging

Goal: prepare a set of strains for learning so that the model becomes “acquainted” with the diversity and range expected for a biological question .



- How many classes does the biological question suggest?
- What do we typically find in each class, based on the literature or experience?
- Are there “extreme” cases?
- How can we balance the classes?



Where the data begin: strains, **biofilms**, and staining – then imaging

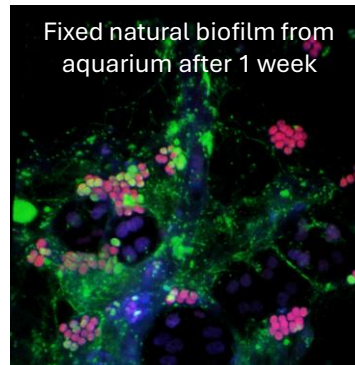
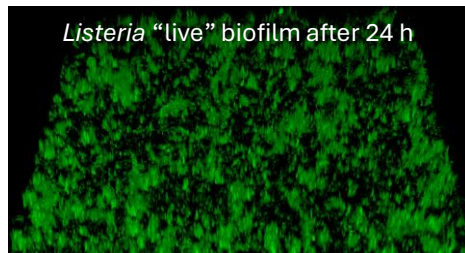
Do your biofilm growth practices need to be adapted for microscopy -> MicroICS?

Target organism(s):

- Natural biofilms
- Laboratory grown biofilms

- Live samples
- Fixed samples

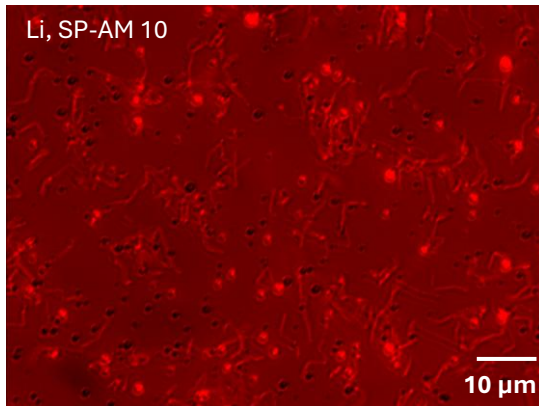
- Materials used to grow biofilms
vs
- **Materials suitable for imaging**



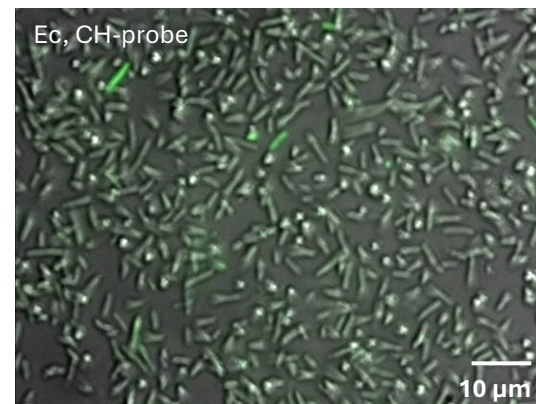
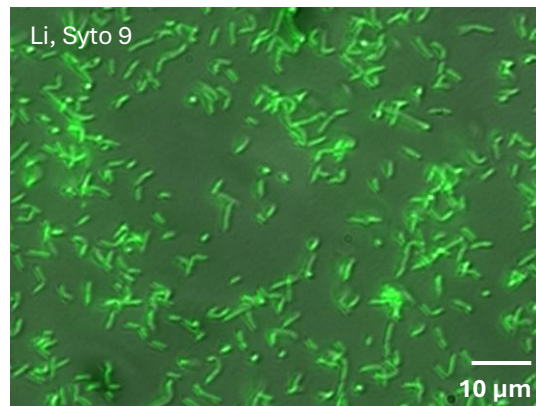
Where the data begin: strains, biofilms, and **staining** – then imaging

Goal: Stain the biofilm in a way that represents the biofilms as you expect.

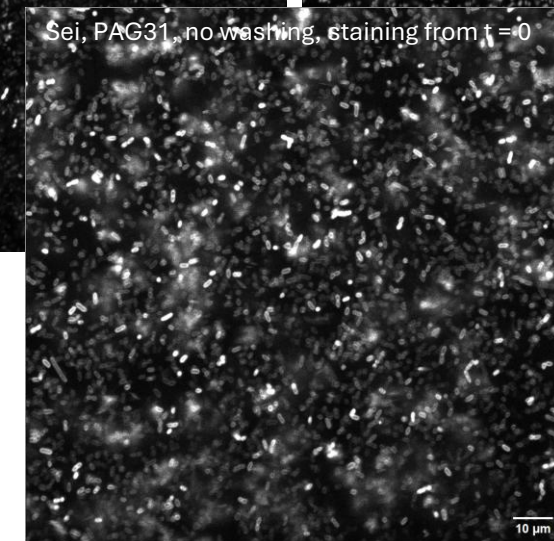
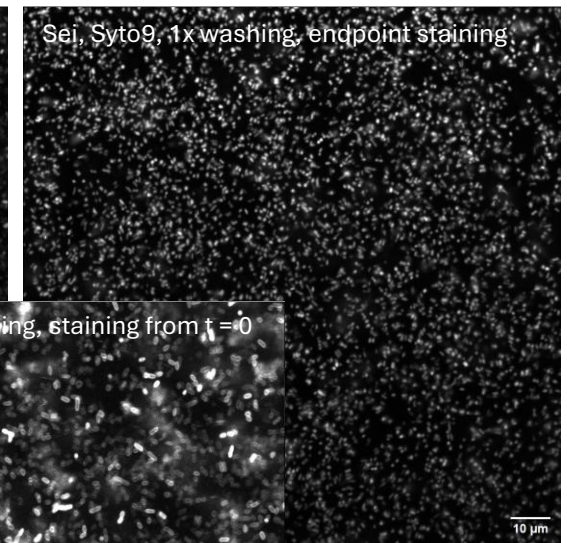
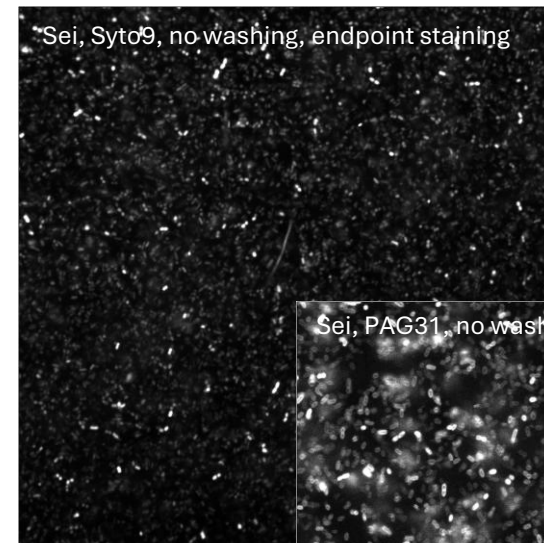
Try different stains. Lack of staining causes lack of information for the model.



Does the stain bind to non-target structures and produce artefacts (washing)?



Does the stain mark all targets in the population?

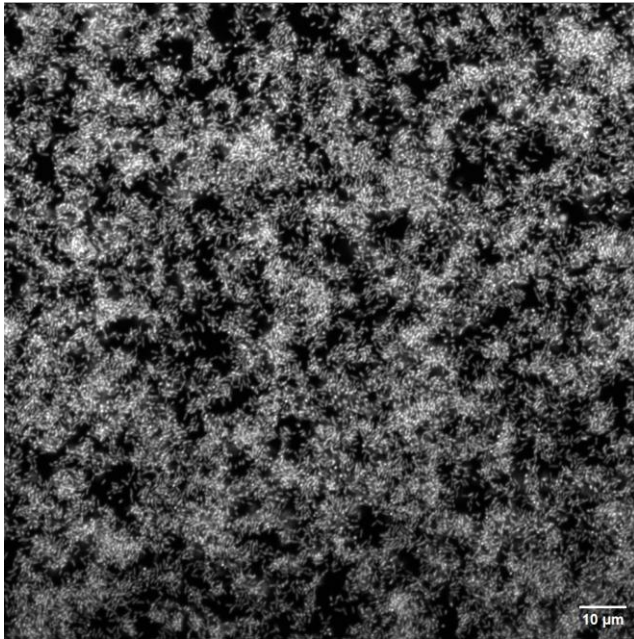


Does the nature of biofilm allow diffusion of the stain throughout the volume for end-point staining?

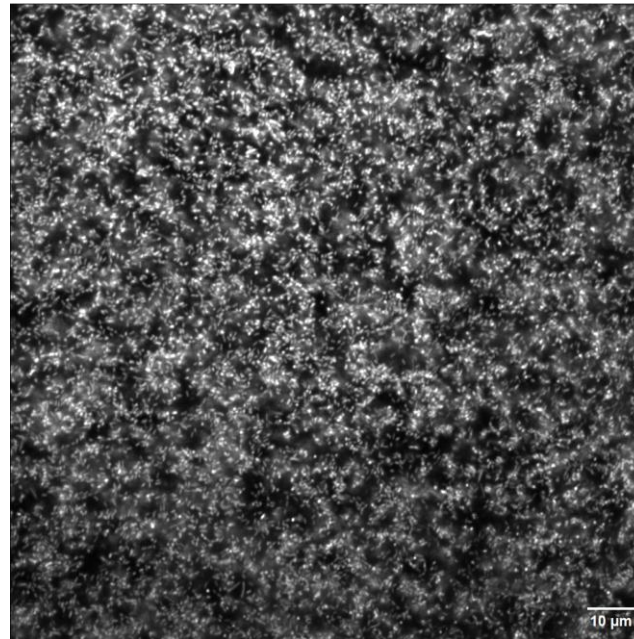
Where the data begin: strains, biofilms, and **staining** – then imaging

Different strains will be stained differently; the worst case must be acceptable.

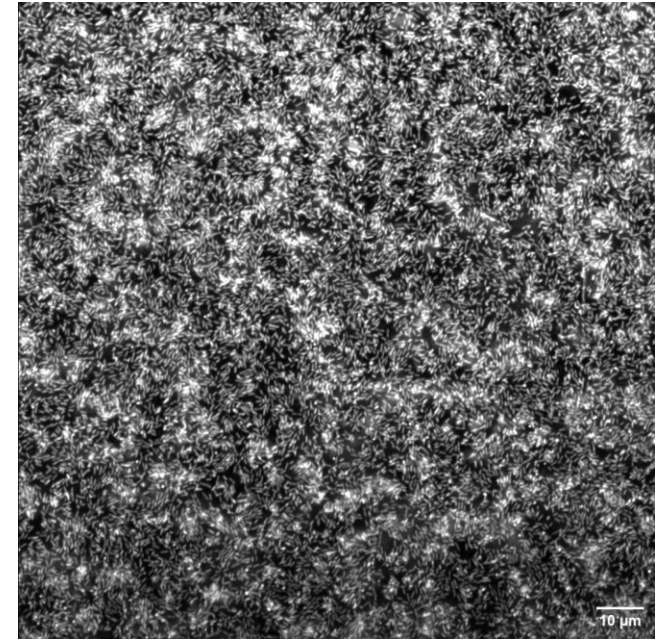
ATCC 19115



L1764

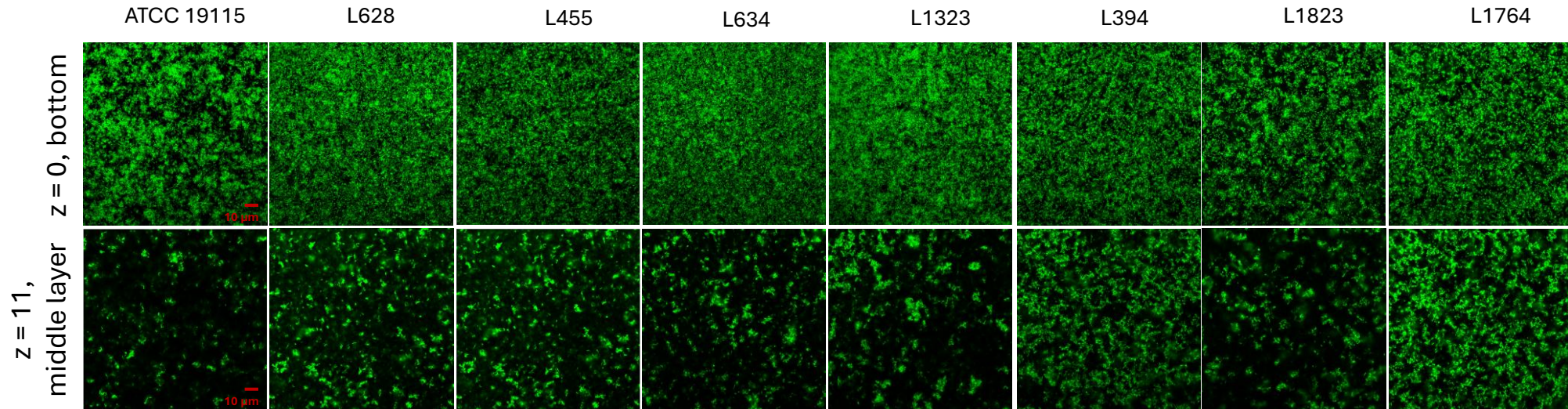


L628



Develop an “understanding” of the biofilms of selected strains to guide image acquisition

- Biofilms are variable – can you estimate visually how much the biofilms differ between the strains (first layer, height, etc.)? How do the biofilms differ within the well, between the wells, and between dates?
- We need to account for natural variability and for variability introduced by the method itself, especially when washing is applied.
- Look for trends rather than special features, unless those features are repeatedly observed.



Microscopy basics and image acquisition

B. Kokot, PhD

Condensed Matter Department, Laboratory of Biophysics

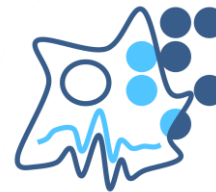
Center for Advanced Optical microscopies (cNOM)

<https://cnom.ijs.si/>

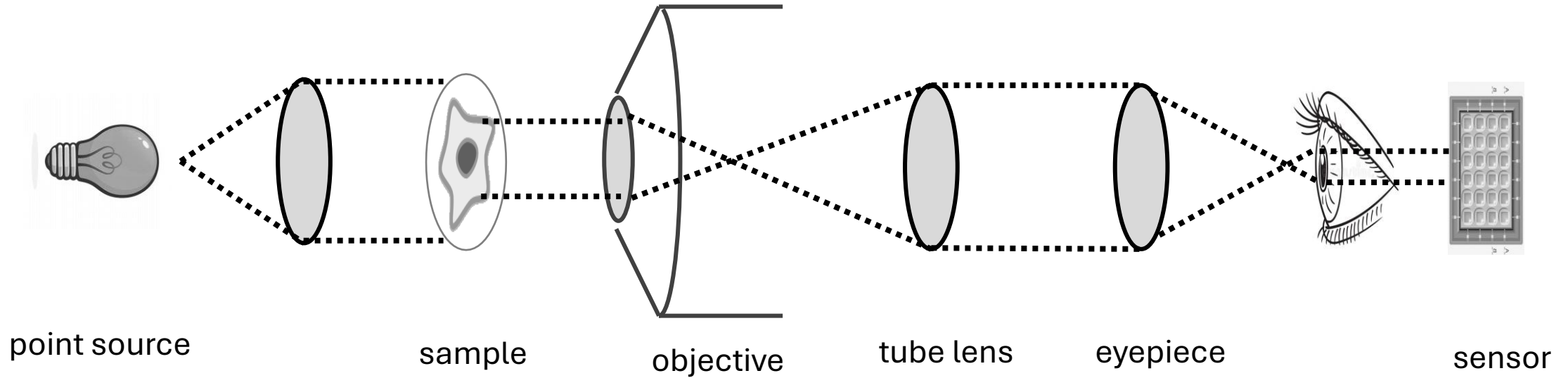


CONDENSED MATTER PHYSICS

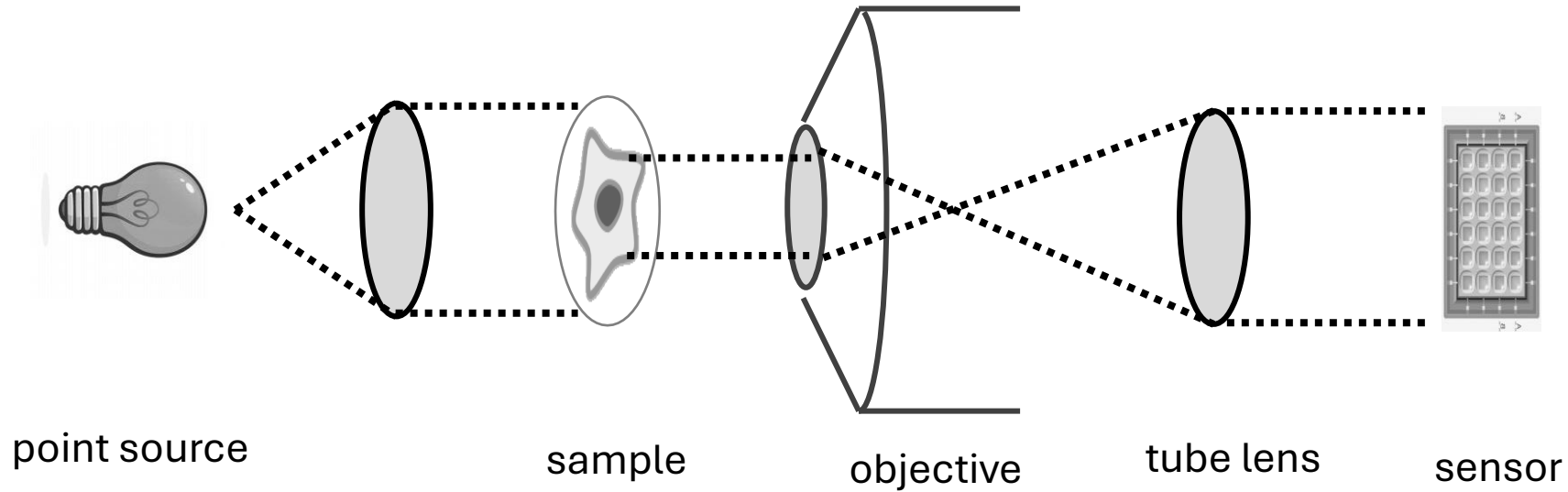
J. Stefan Institute, Ljubljana, Slovenija



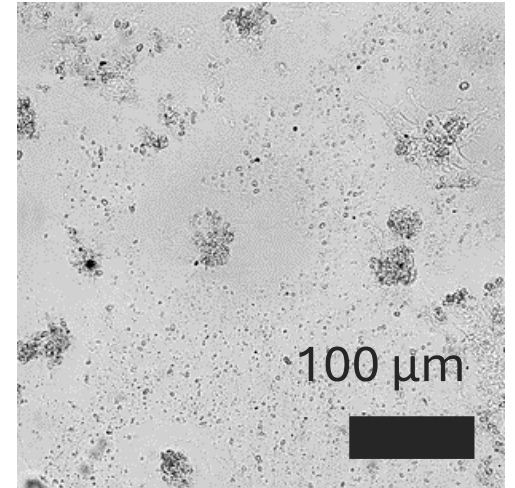
Basic microscope setup



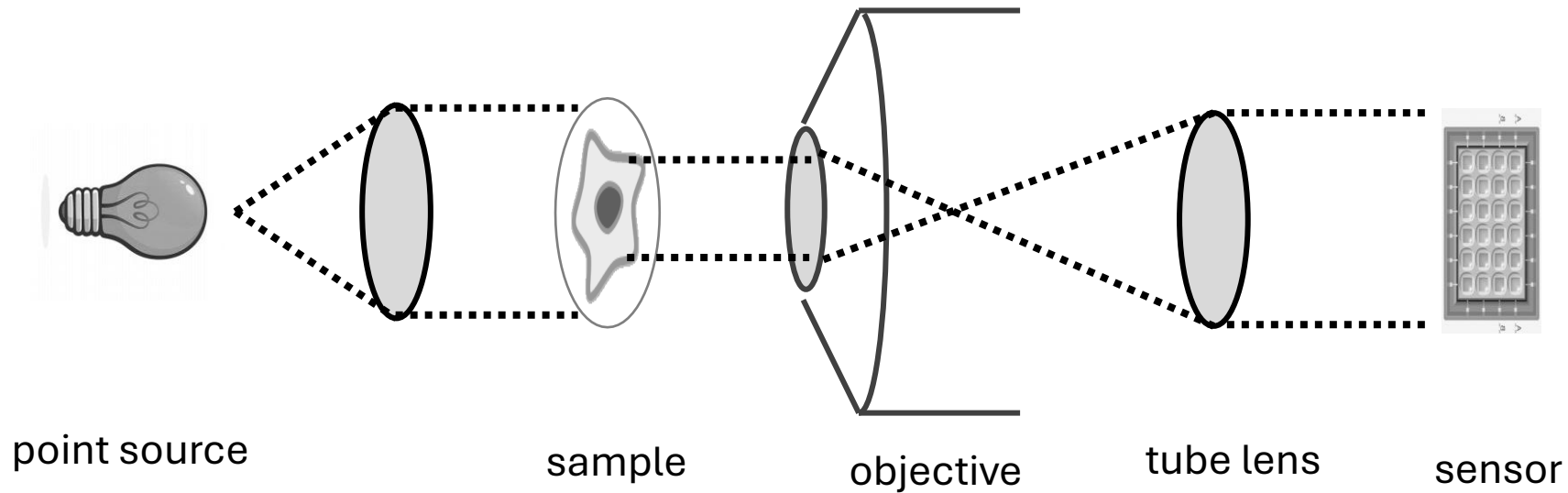
Basic microscope setup



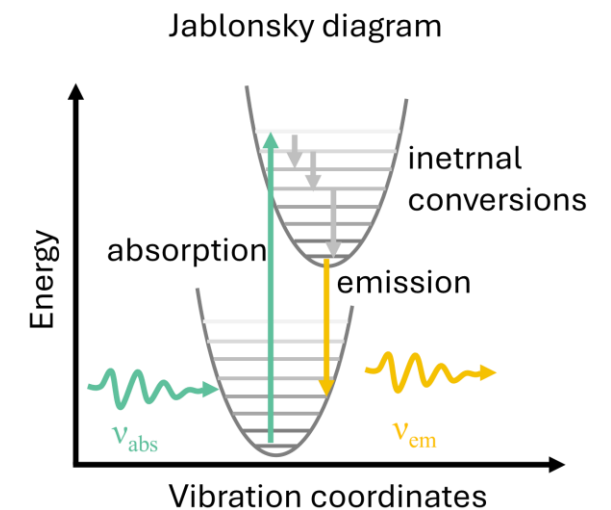
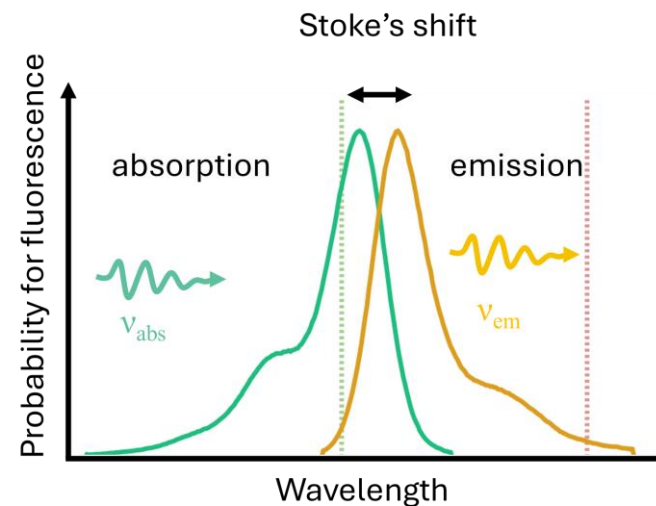
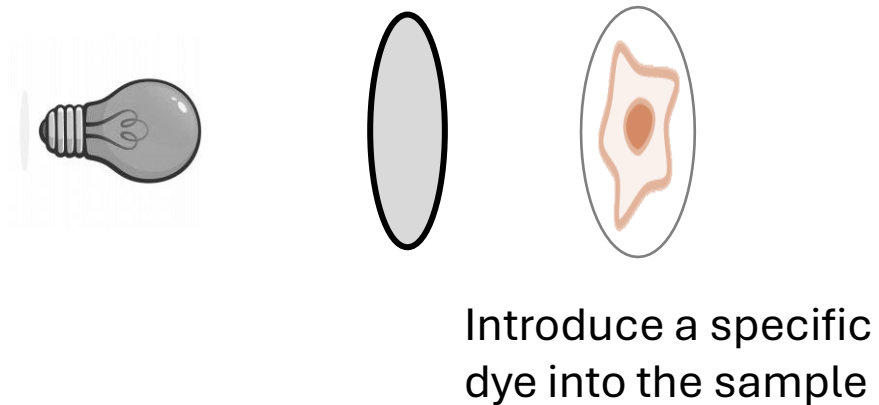
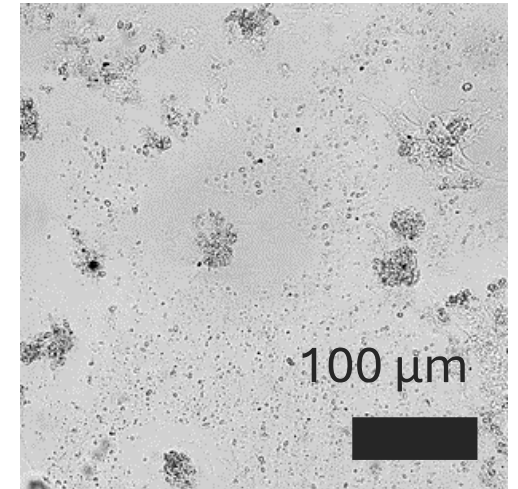
Brightfield



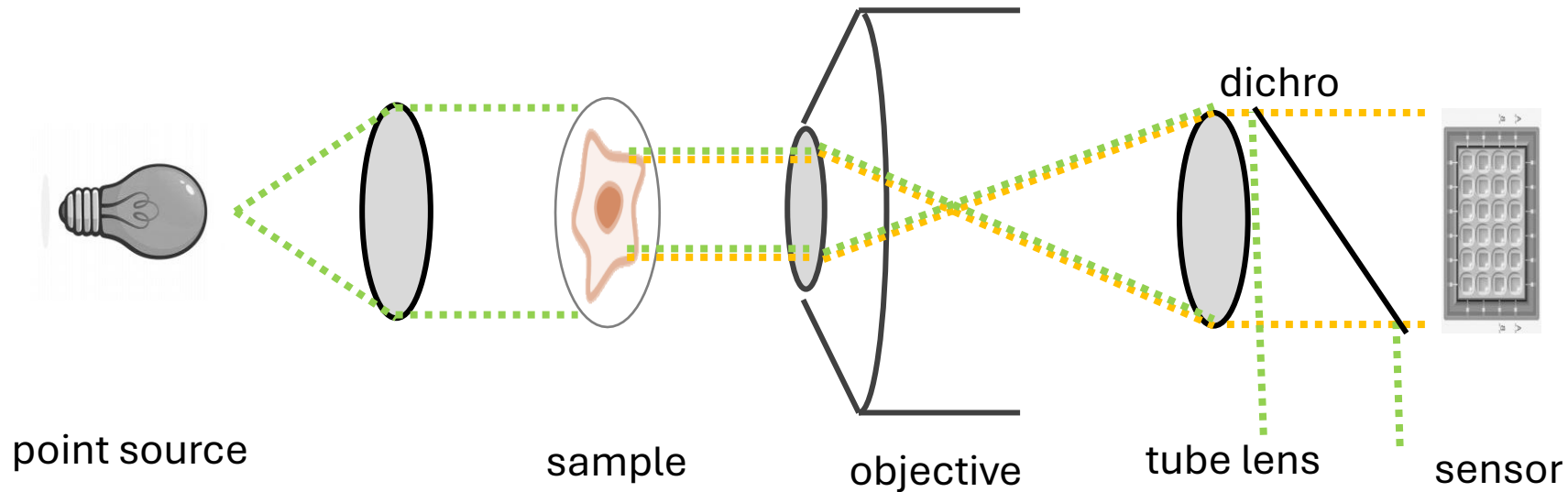
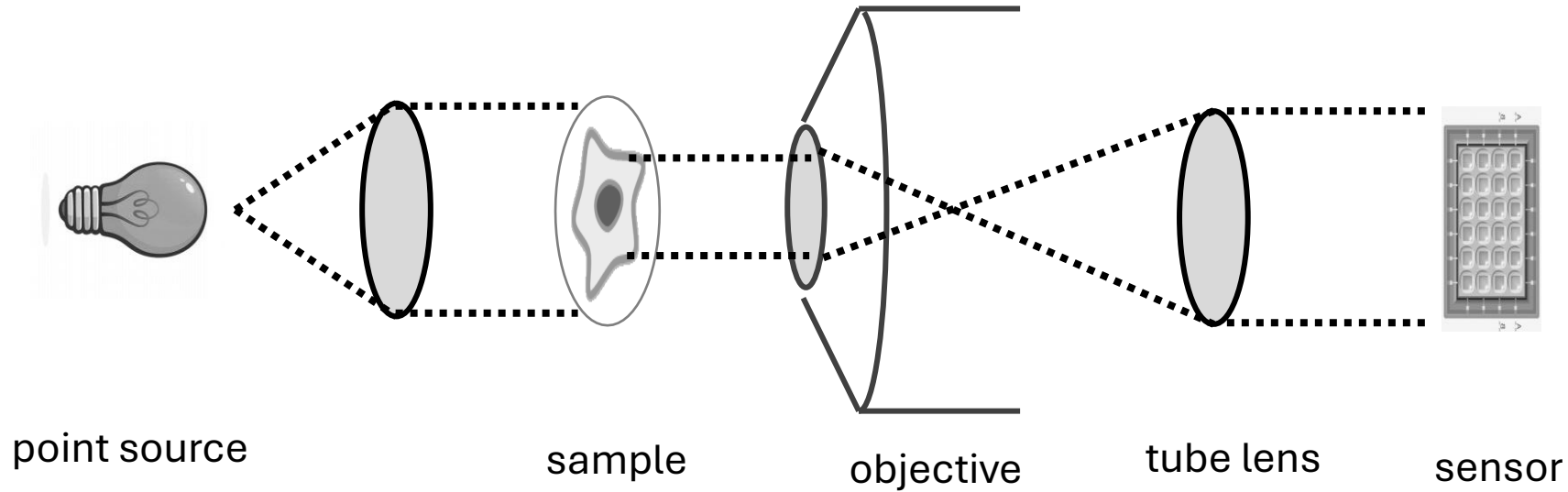
Basic microscope setup



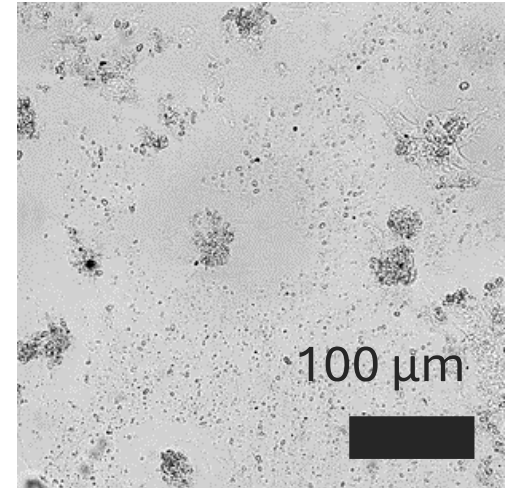
Brightfield



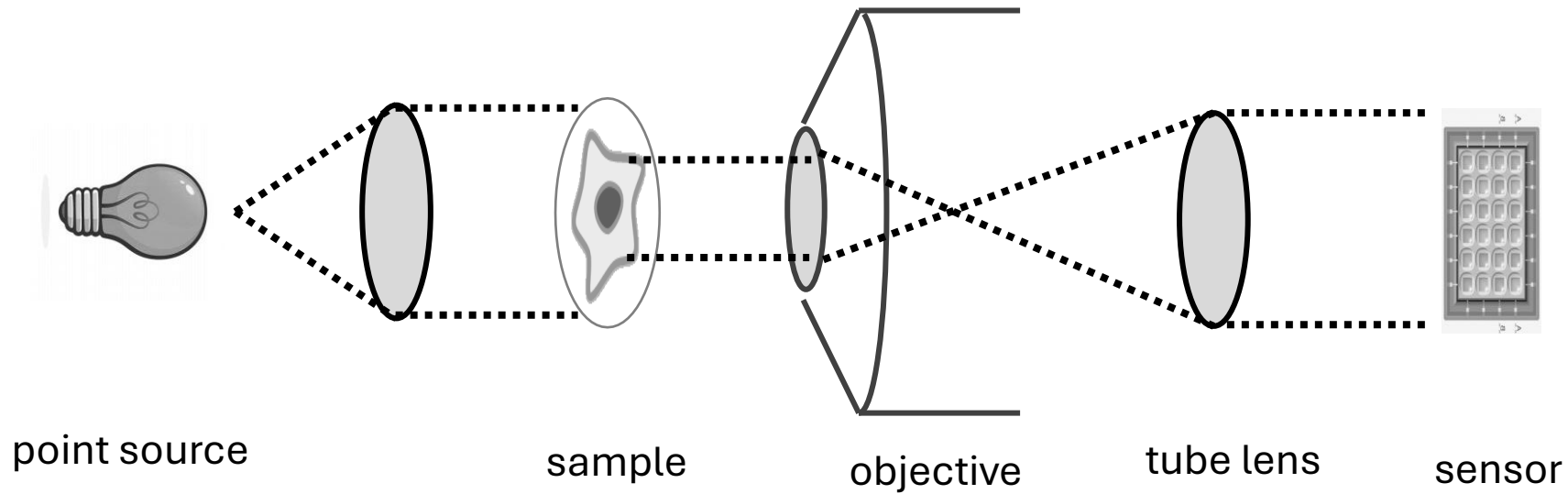
Basic microscope setup



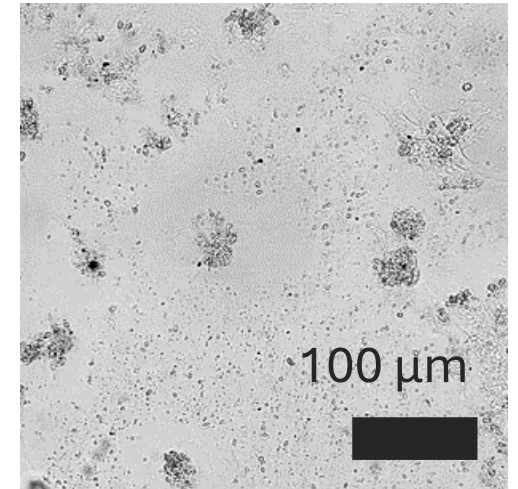
Brightfield



Basic microscope setup



Brightfield



cell membranes + nanoparticles

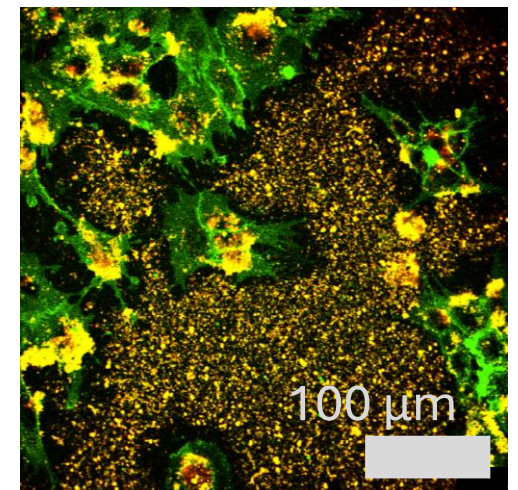
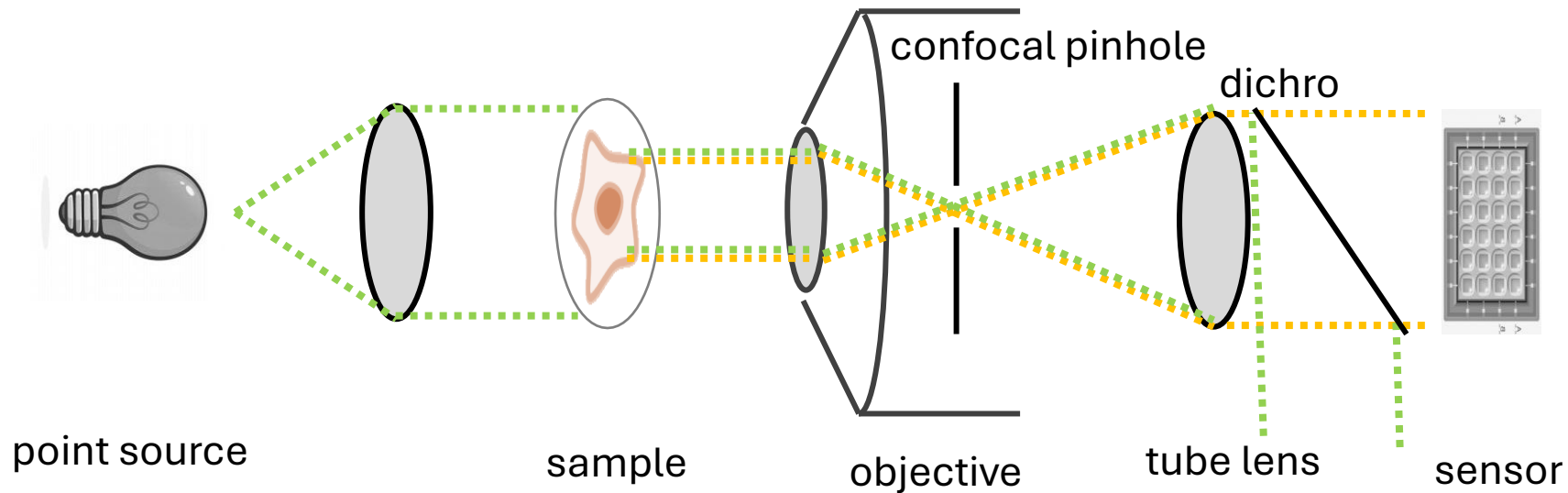
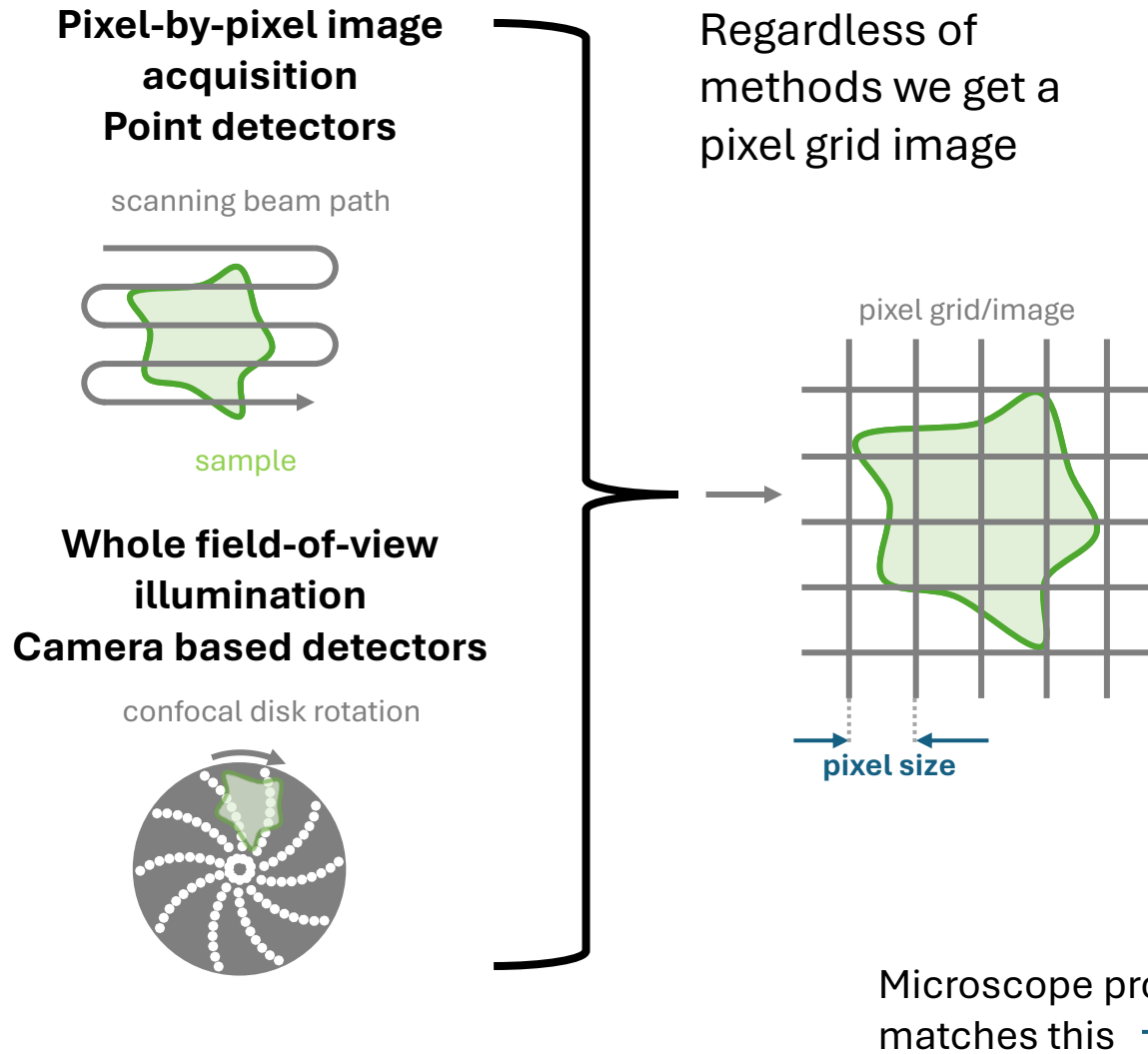
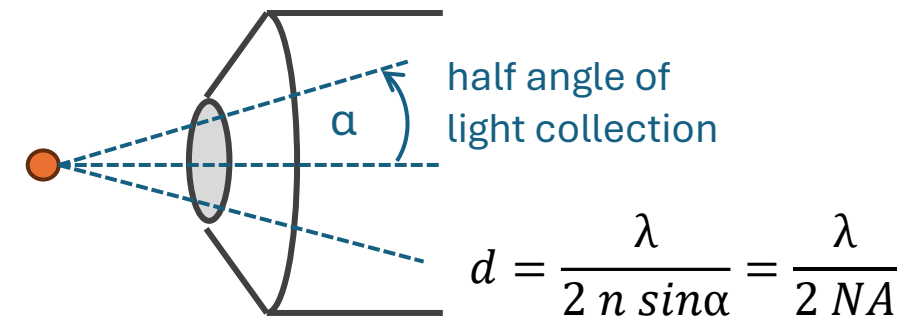


Image formation



How to determine pixel size?

- Widefield microscopy = predefined by microscope provider
- Scanning microscopy = chosen by a user. But how to choose?



d = resolution limit

n = refractive index of immersion media

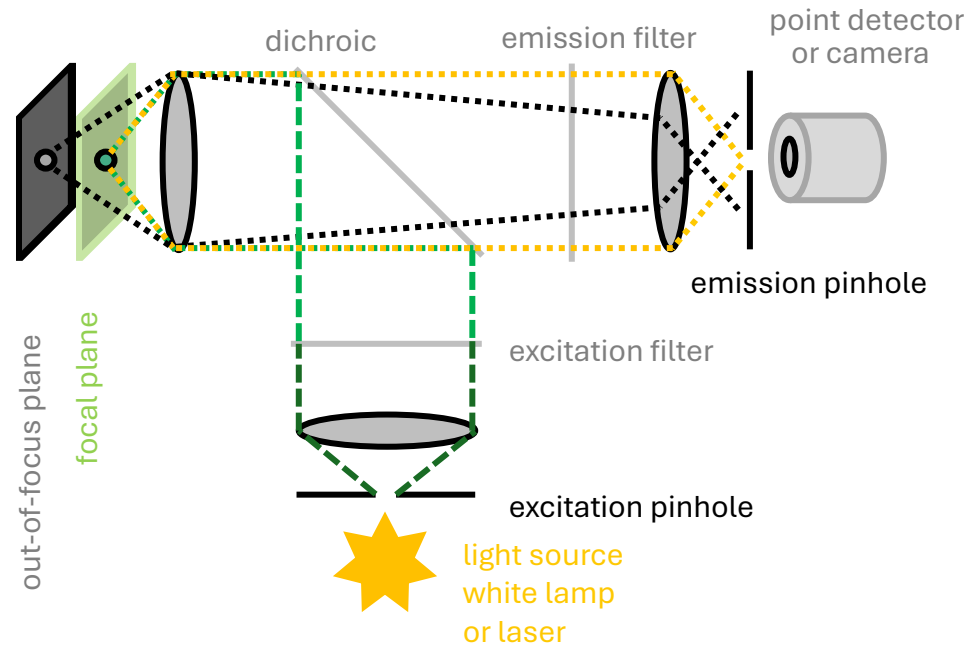
NA = numerical aperture of the objective

λ = wavelength of incident light

Microscope provider matches this

Rule of thumb (Nyquist sampling): $pixel\ size = \frac{d}{2}$

Acquiring z-stacks



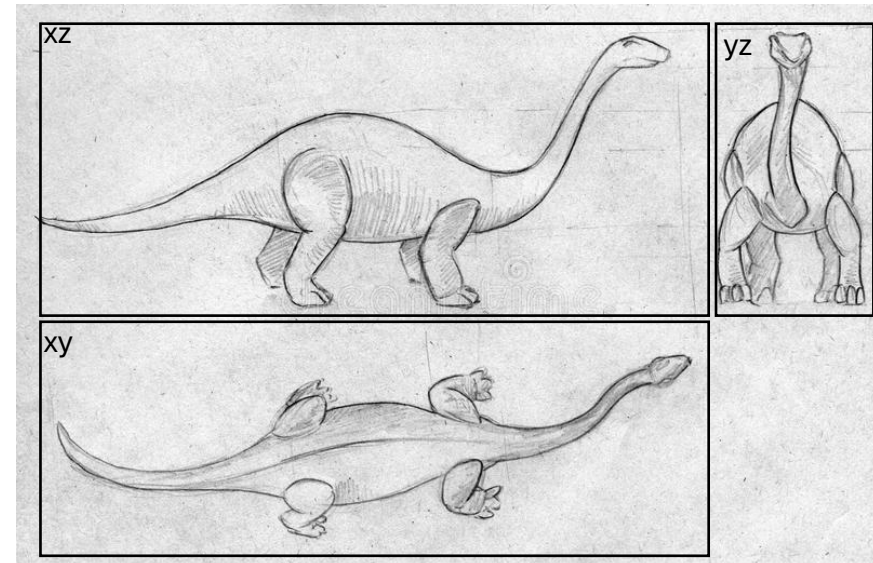
$$d_{xy} = \frac{\lambda}{2 n \sin \alpha} = \frac{\lambda}{2 NA}$$

$$d_z = \frac{2 \lambda}{NA^2}$$

Rule of thumb (Nyquist sampling): $pixel\ size = \frac{d}{2}$

How to determine pixel size?

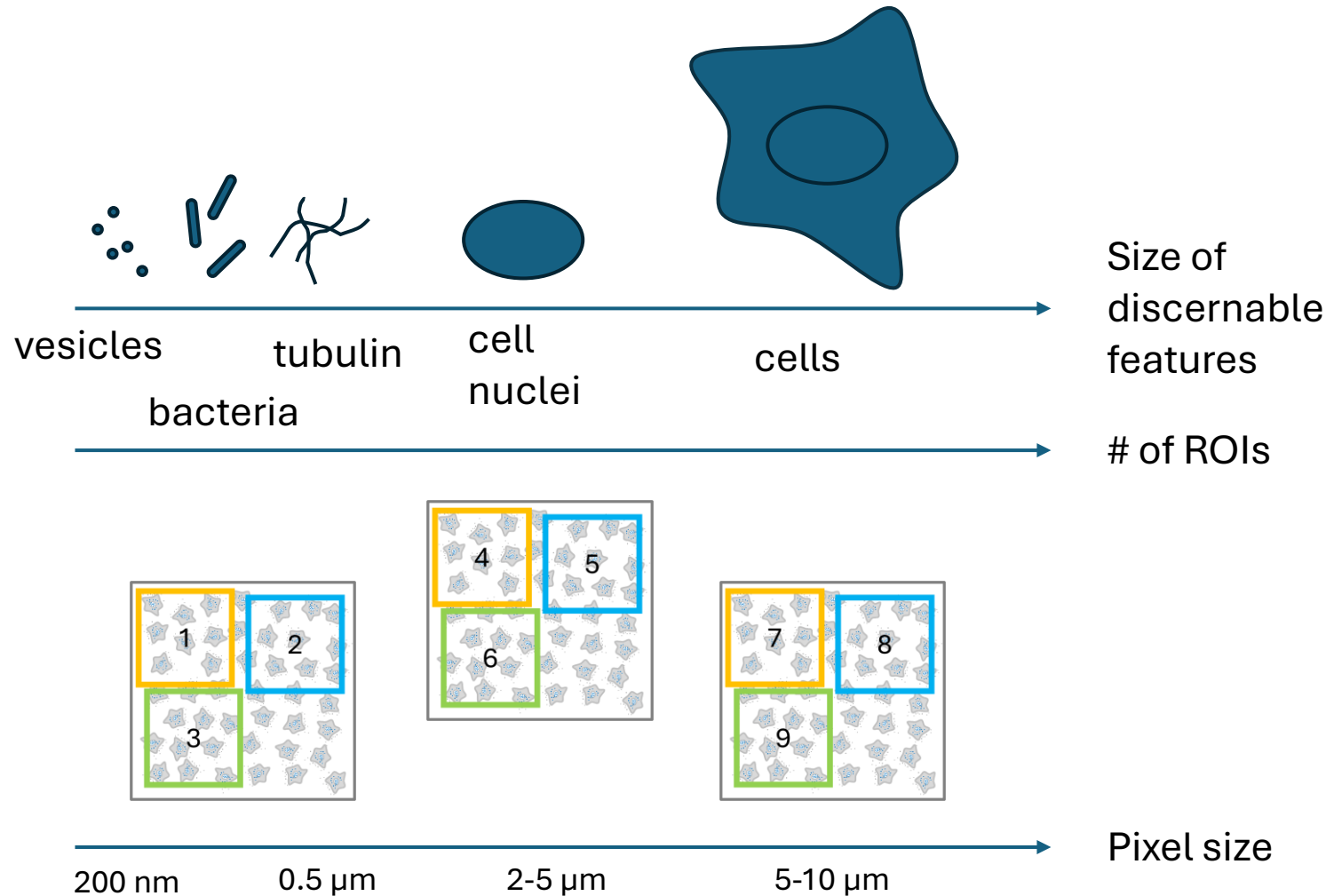
- You have to have good enough sampling for the structures you want to observe
- 1 μm big structures would require 0.5 μm pixel size (following Nyquist sampling)



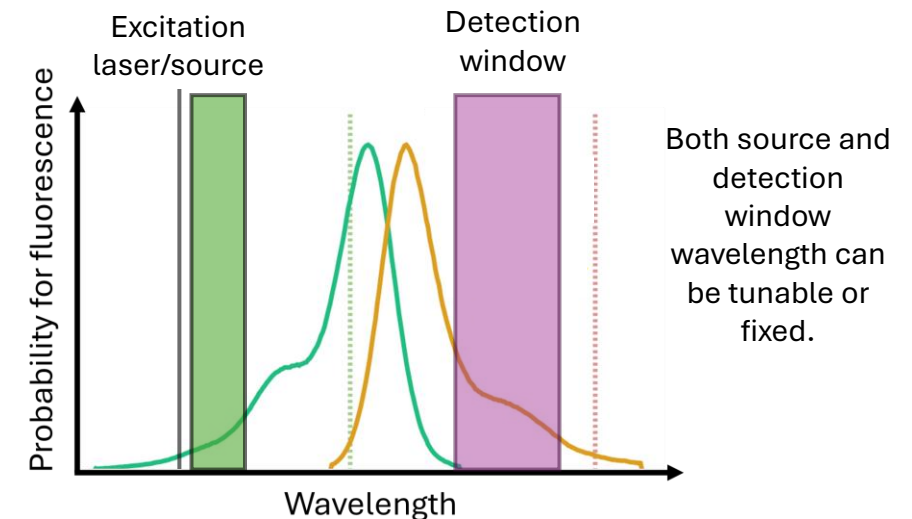
How to optimize (and compromise)

- Determine the structure size of interest
- Sample finely enough to capture those structures
- If unsure sample more finely, as to coarse sampling and microICS analysis will surely fail
- How much statistics do I need? Do I care about the speed of acquisition?
 - Acquire just enough so the data is relevant. Less acquisition time means more recorded samples.
- Record all representative parts of the sample in the observed representative ratios.
- Optimizing speed of acquisition
 - Determine the largest pixel size you can afford
 - Determine maximal ROI (FOV) size your microscope allows at chosen pixel size and the objective
 - Check for drifts during one ROI acquisition
 - Time one ROI
 - Determine how many replicates do you wish to record in a chosen time window
 - Divide time window with time for single ROI and distribute the number of FOVs between replicates

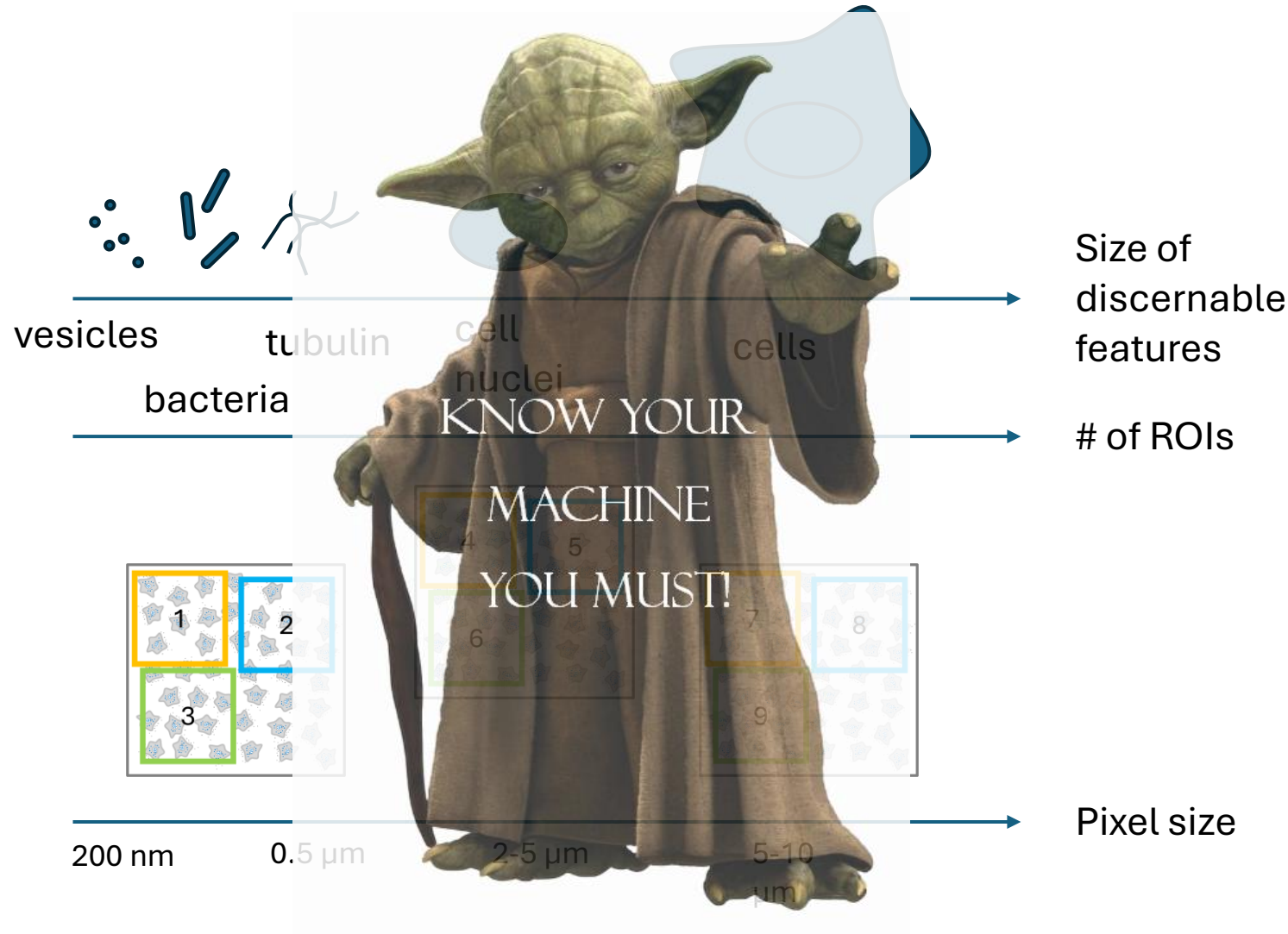
How to optimize (and compromise)



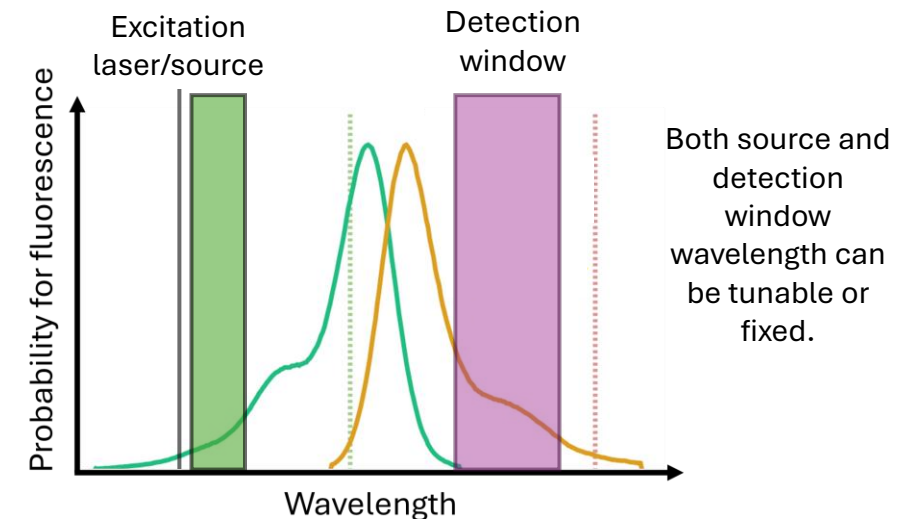
- Additional things to consider on the detection side
 - What is a good enough signal? What powers to use? How to set gain?



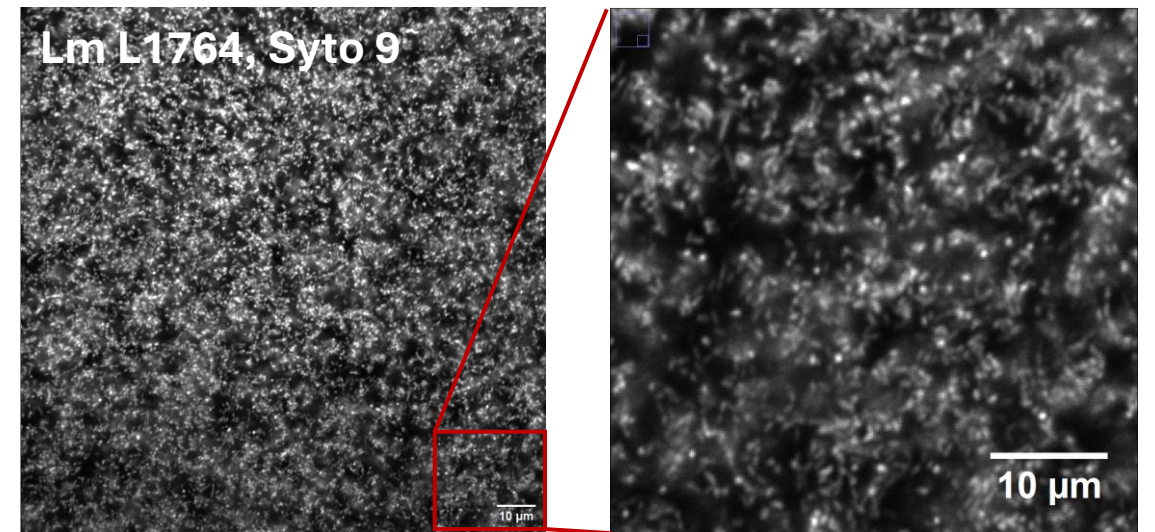
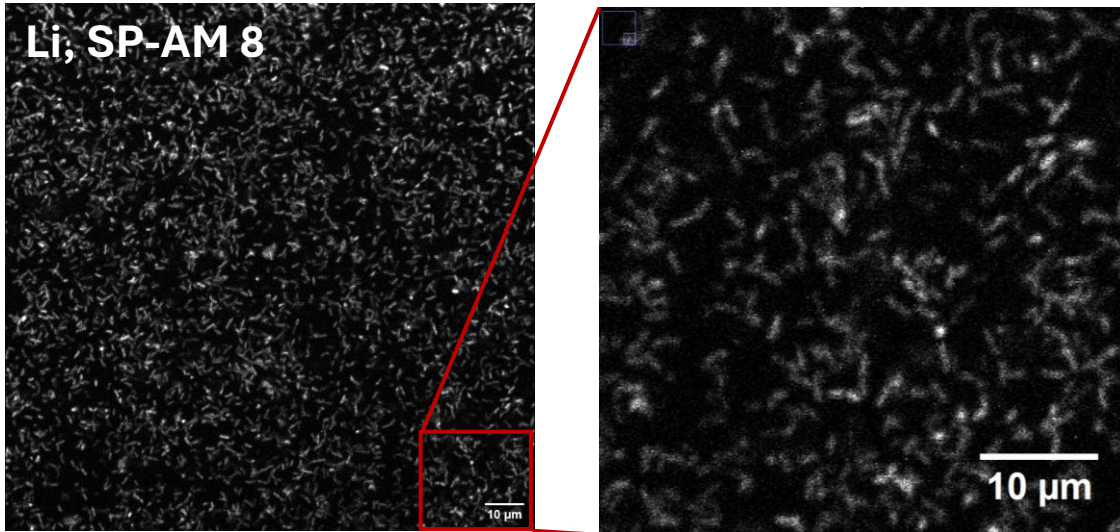
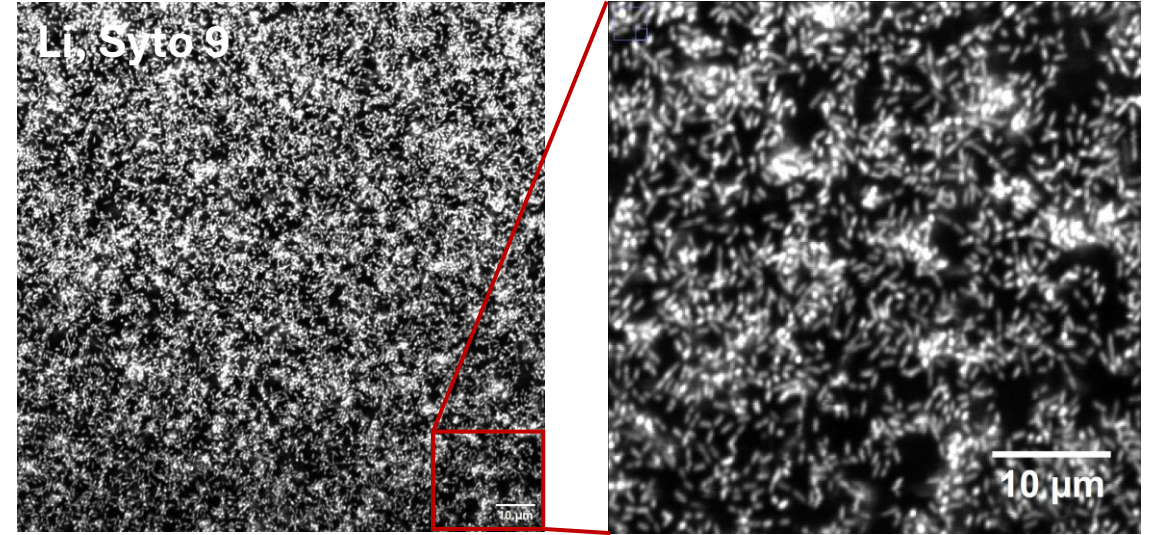
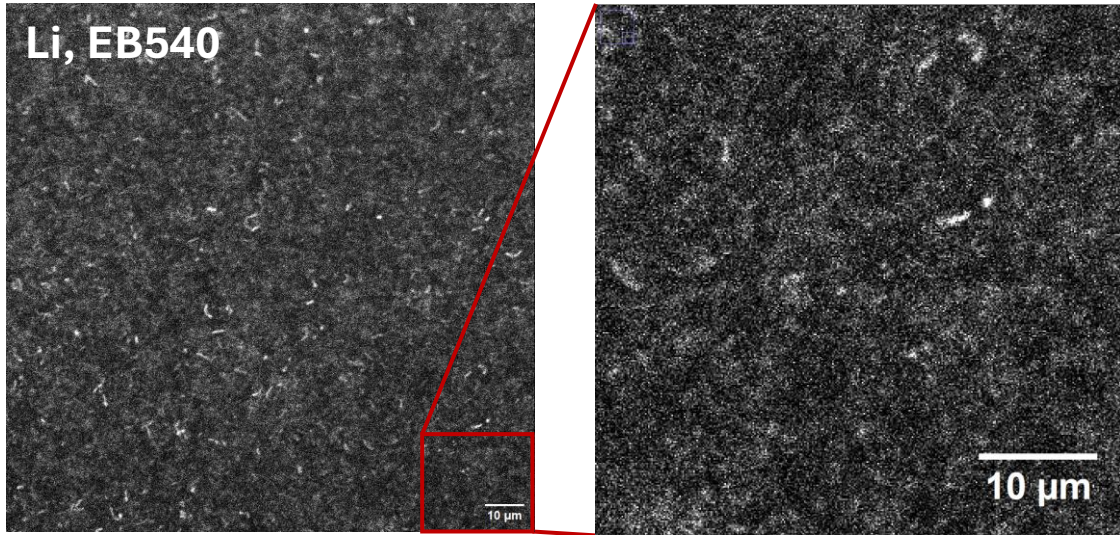
How to optimize (and compromise)



- Additional things to consider on the detection side
 - What is a good enough signal? What powers to use? How to set gain?



Quick signal-to-noise ratio example





Run MicroICS + Read and interpret



Funded by
the European Union

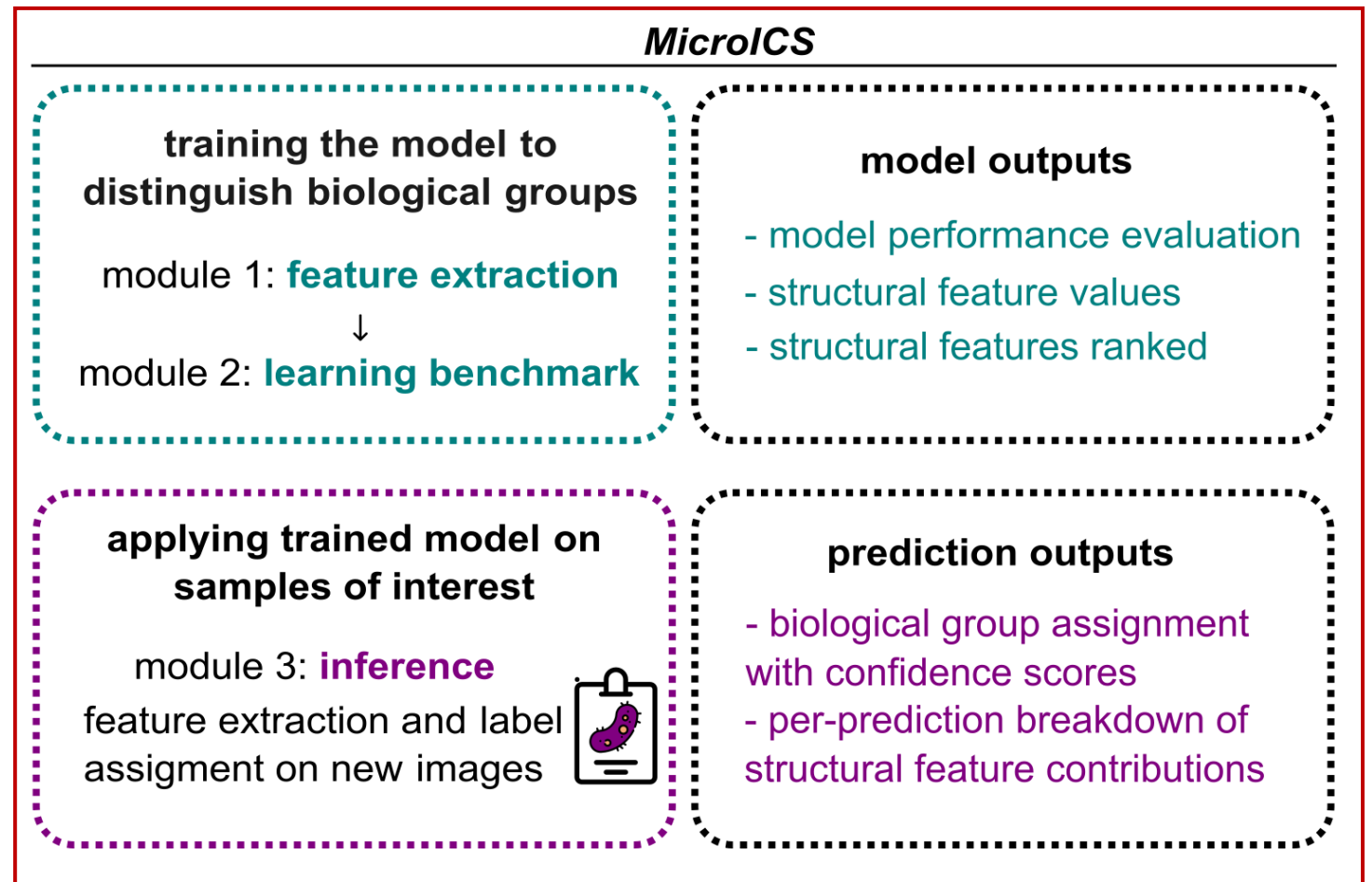
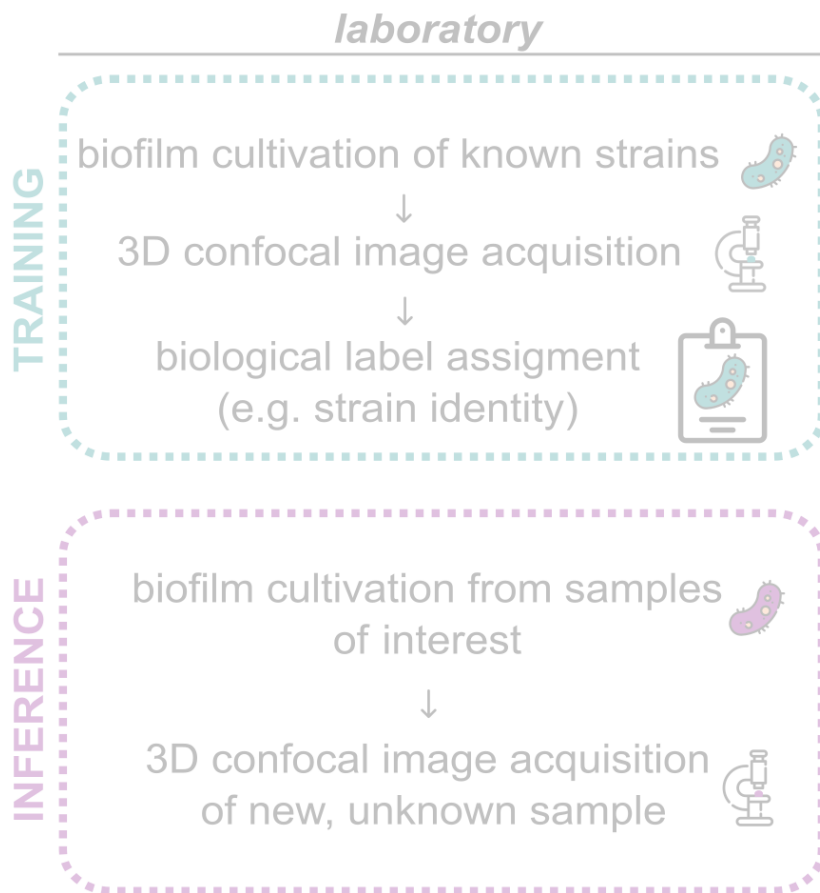


Biofilm
Regulatory
Toolbox



Jožef Stefan
Institute
Ljubljana, Slovenia



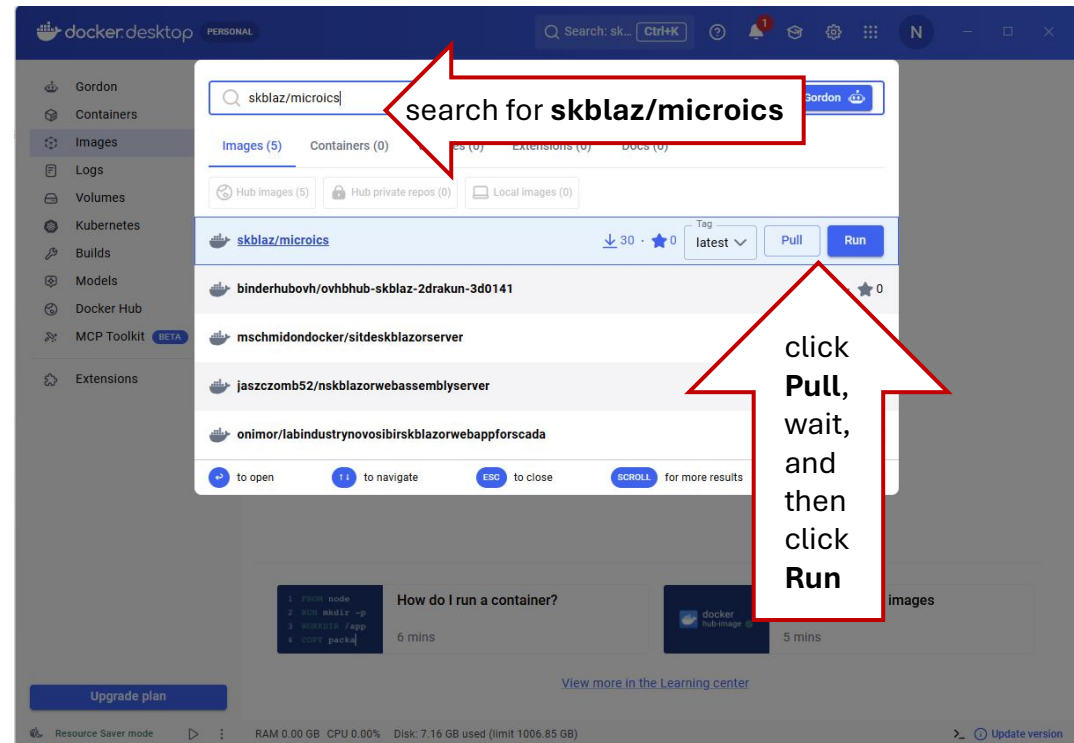


Installation of MicroICS

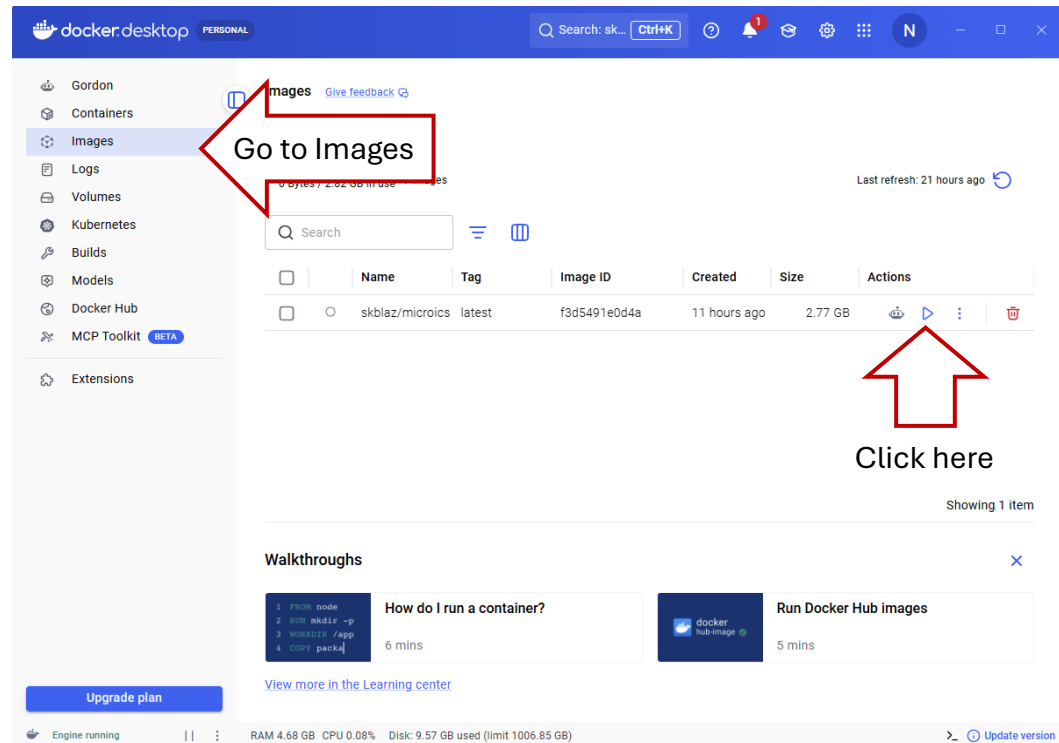
Document “Installation guide”

Did you install Docker Desktop?

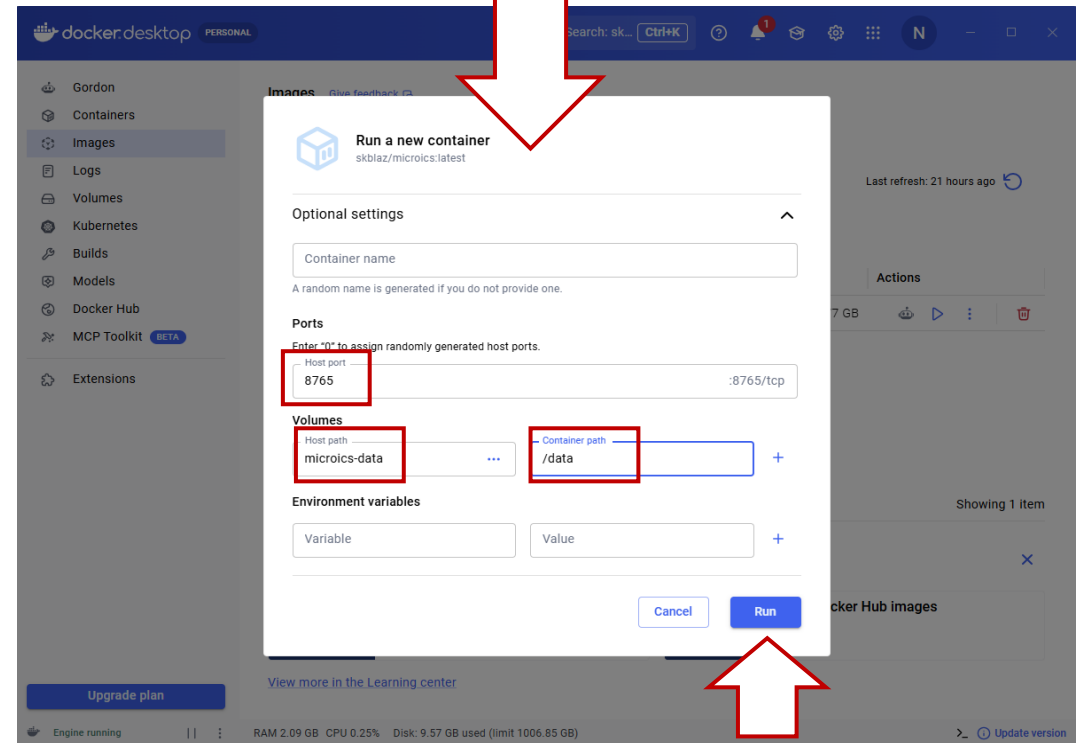
Wi-fi: guest
Password: dearguest



Installation of MicroICS

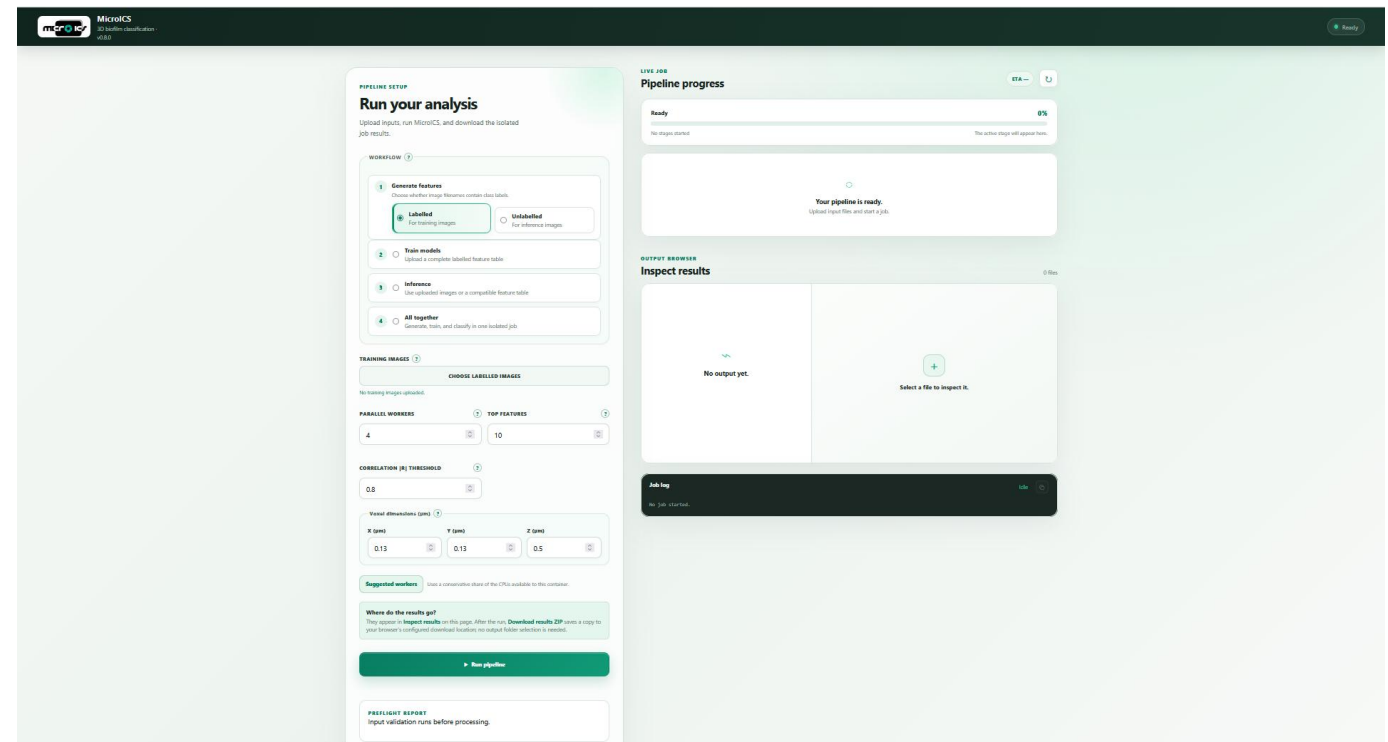
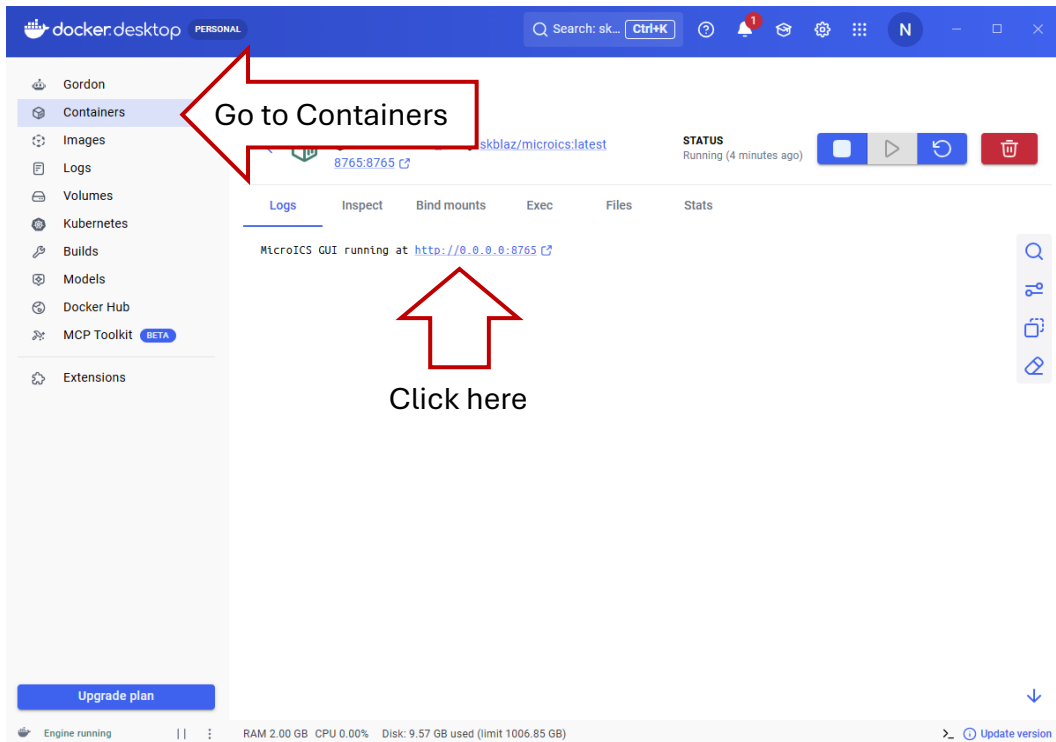


Type in information in red boxes



Click here

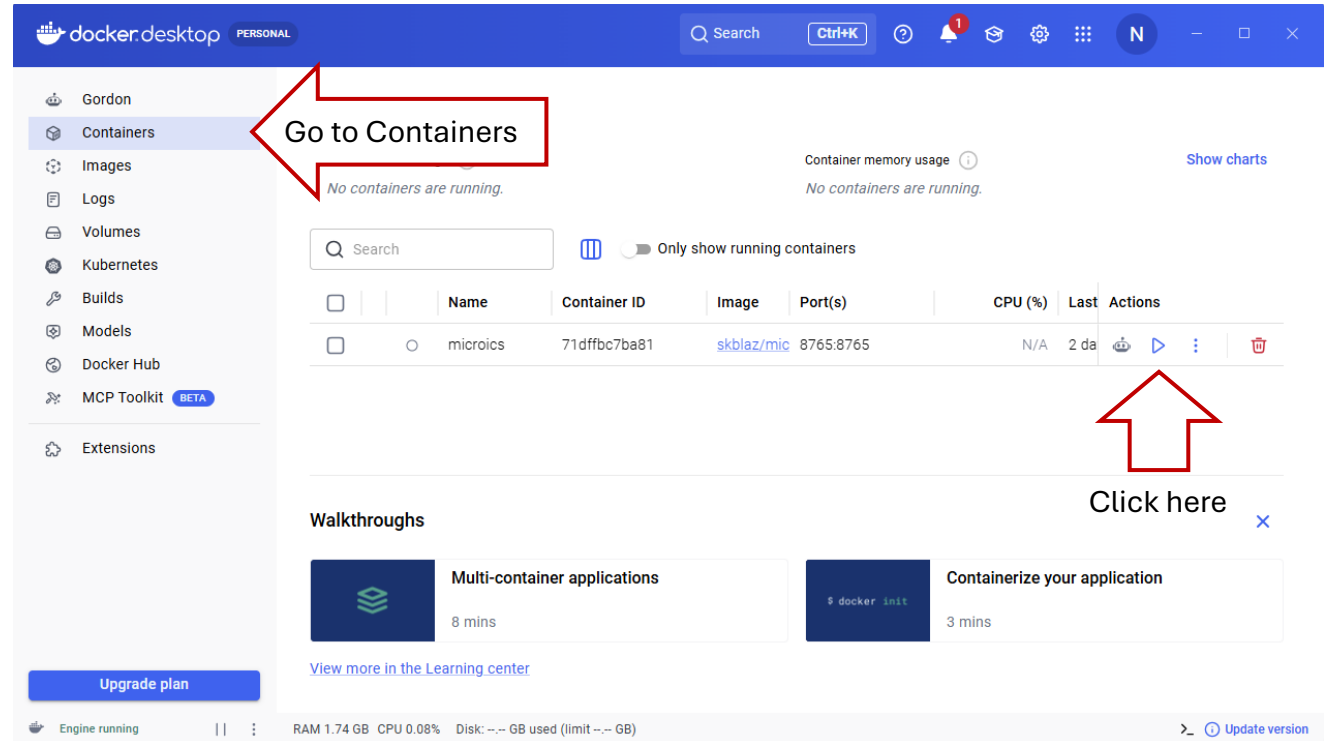
Installation of MicroICS



Installation of MicroICS

If you installed MicroICS:

- Start Docker Desktop.
- Go to Containers.
- Click Run.
- Click the link.



☐	Name	Container ID	Image	Port(s)	CPU (%)	Last	Actions
☐	microics	71dffbc7ba81	skblaz/mic	8765:8765	N/A	5 se	   

Click here

Sample data

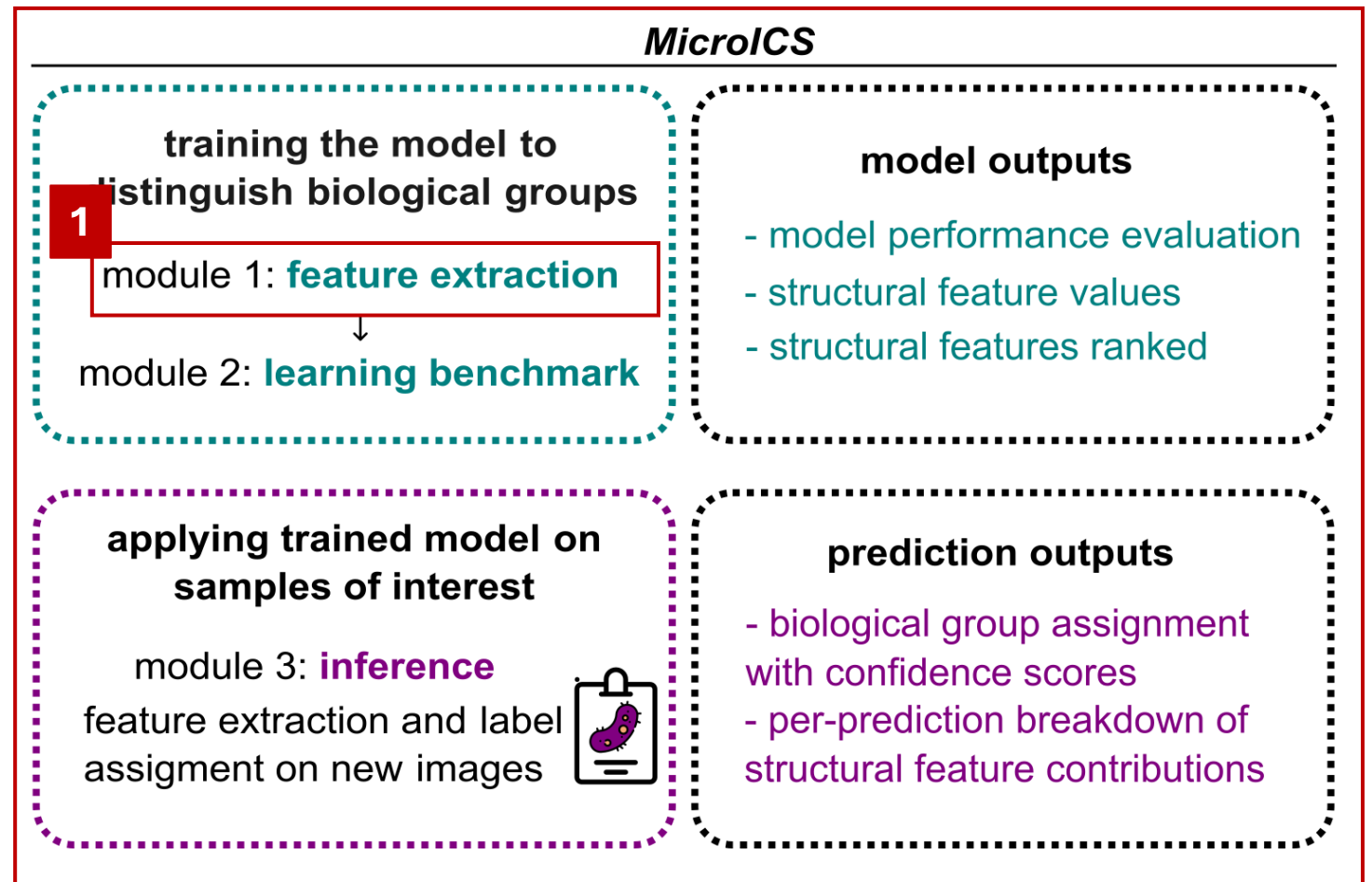
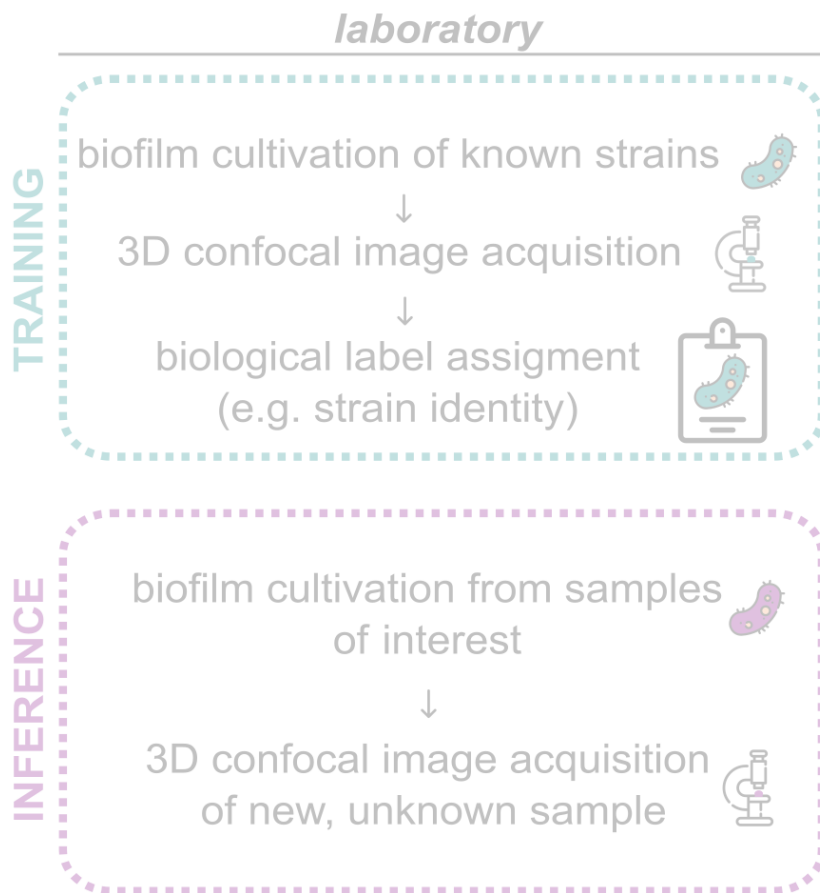
We need these folders:

Click on:

The screenshot shows the Google Cloud Storage web interface. On the left, a sidebar lists folders under 'COST_TS_materials_image_analysis'. The main area displays a table of files. A red box highlights the three-dot menu icon next to the first file, and another red box highlights the 'Download' option in the dropdown menu.

Name	Size	Modified
images_generate_features_labelled	336 MB	22 days ago
images_generate_features_unlabelled	336 MB	22 days ago
inference_clinical_assoc	82.4 MB	2 hours ago
inference_identity	183.3 MB	2 hours ago
trained_models_clinical_assoc	305.3 MB	2 hours ago
trained_models_identity	101.5 MB	2 hours ago

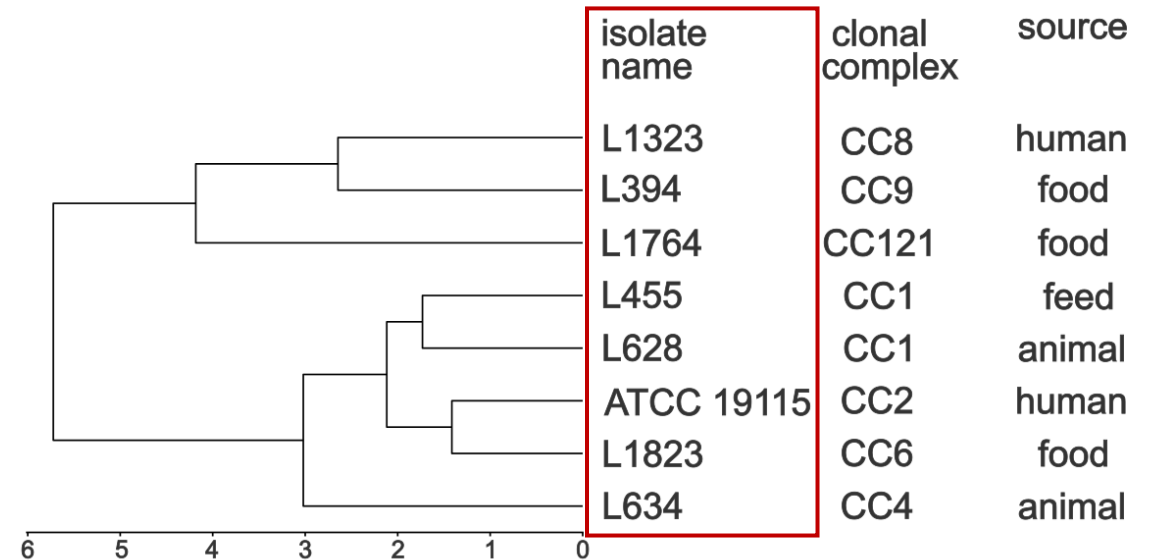
6 folders 1.3 GB



Our first analysis: training the model to distinguish between strains

Remember?

We prepared 1000 images of 8 strains (125 images per strain), and we named the images so that the strain name (label) is evident from the name.



Our first analysis: generate features

Data (unzip):

<https://portal.ijs.si/nextcloud/s/LC93n9A97Ekme7X>

Folder:

images_generate_features_labelled

The image shows a web interface for generating features, with five numbered callouts highlighting specific elements:

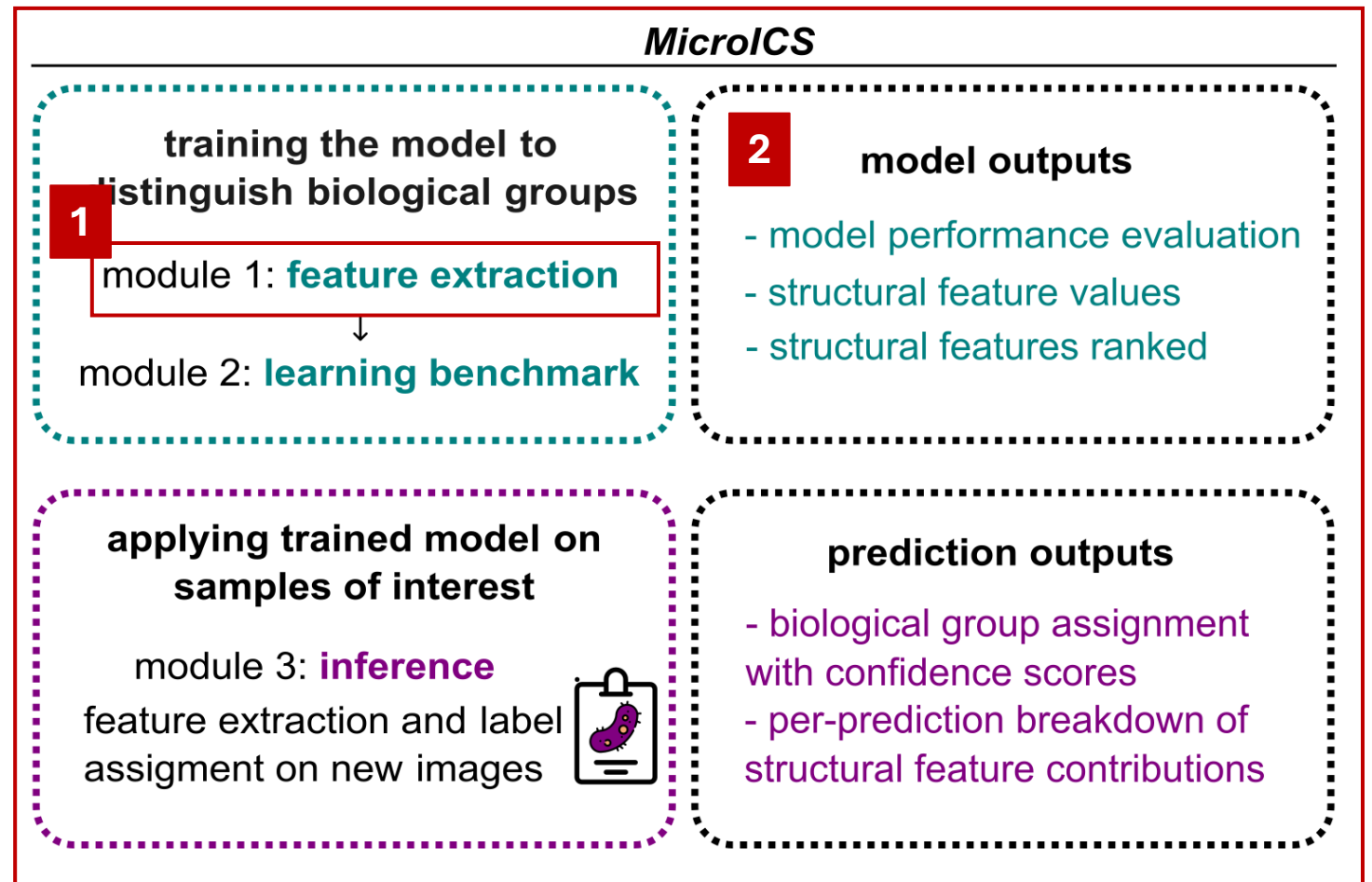
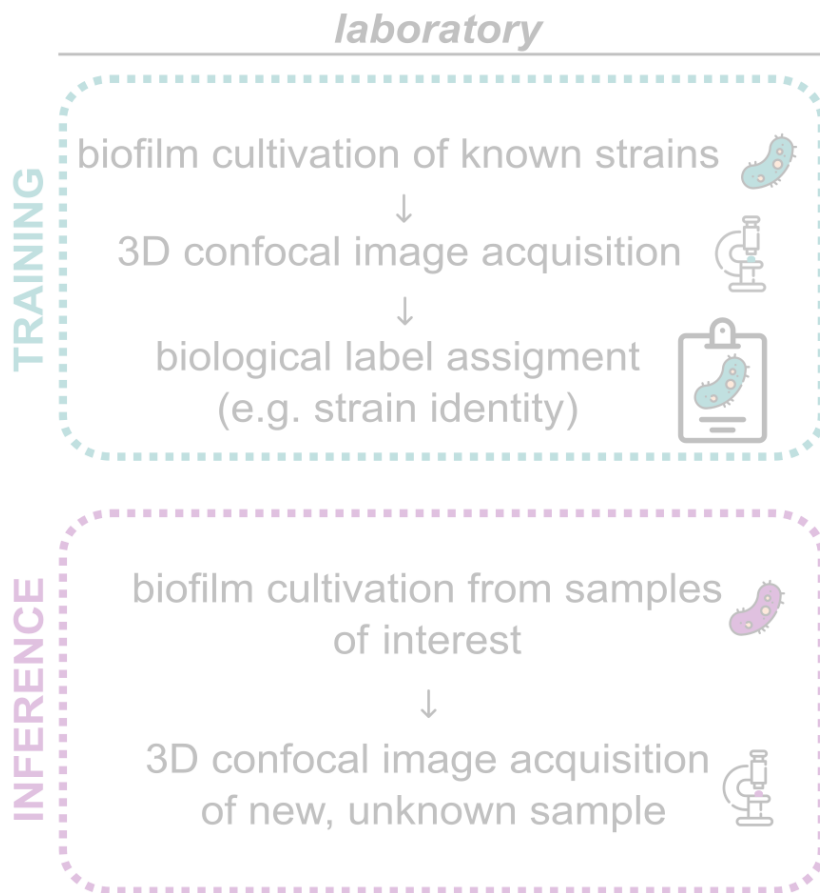
- 1**: Points to the 'Generate features' section, which includes a sub-section 'Choose whether image filenames contain' with two radio buttons: 'Labelled' (selected, for training images) and 'Unlabelled' (for inference images).
- 2**: Points to the 'TRAINING IMAGES' section, which includes a 'CHOOSE LABELLED IMAGES' button and a message 'No training images uploaded.'
- 3**: Points to the 'PARALLEL WORKERS' and 'TOP FEATURES' sections. 'PARALLEL WORKERS' has a dropdown set to 4. 'TOP FEATURES' has a dropdown set to 20.
- 4**: Points to the 'CORRELATION |R| THRESHOLD' section, which has a dropdown set to 0.8.
- 5**: Points to the 'Run pipeline' button, which is a green button with a right arrow.

Other visible elements include:

- 2**: 'Train models' section with a radio button and the instruction 'Upload a complete labelled feature table'.
- 3**: 'Inference' section with a radio button and the instruction 'Use uploaded images or a compatible feature table'.
- 4**: 'All together' section with a radio button and the instruction 'Generate, train, and classify in one isolated job'.
- 4**: 'Voxel dimensions (μm)' section with three dropdowns for X (μm), Y (μm), and Z (μm), all set to 0.13, 0.13, and 0.5 respectively.
- 5**: 'Suggested workers' section with a note 'Uses a conservative share of the CPUs available to this container.'
- 5**: 'Where do the results go?' section with a note 'They appear in **Inspect results** on this page. After the run, **Download results ZIP** saves a copy to your browser's configured download location; no output folder selection is needed.'
- 5**: 'PREFLIGHT REPORT' section with the text 'Input validation runs before processing.'

Our first analysis: generate features

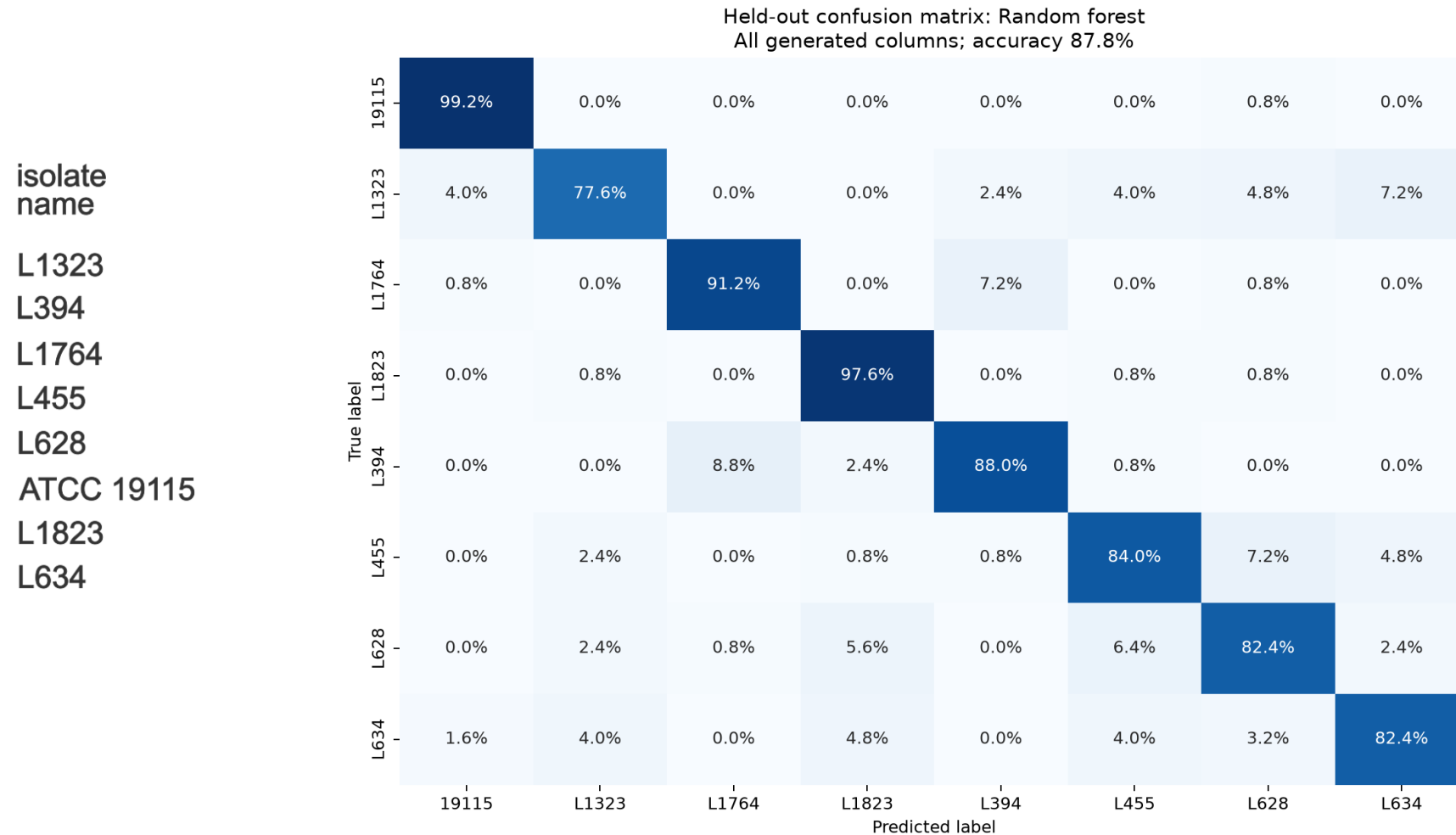
- Download the results.
- Open the file datafile.tsv by dragging it into an open Excel sheet.
- Most of the columns in that table are features – structural properties measured in the same way for every image.
- The names are verbose, machine-generated and numerous (approximately 2,700), and you are not expected to understand them all in advance.
- The point of the tool is the opposite: let the model decide which ones capture meaningful aspects, and then spend your time understanding those in the context of your dataset.



Our first analysis: train models

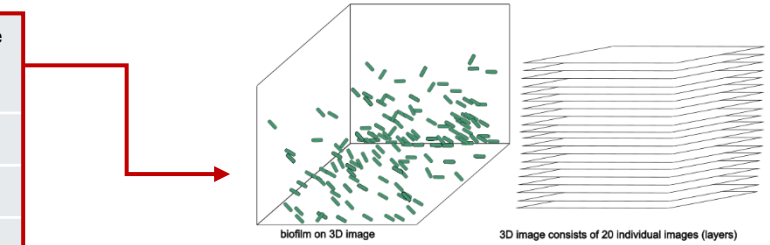
- Training the model to output meaningful results takes too long, so we provide a pre-made model for the training school and do not spend the session waiting.
- Thus, skip this part for now, but not for the homework.
- The pre-made models are available here: <https://portal.ijs.si/nextcloud/f/53497958>, folder **trained_models_identity**
- **We will look at the outputs – refer back in your memory to the theory of ML!**

Our first analysis: what the trained model reports.

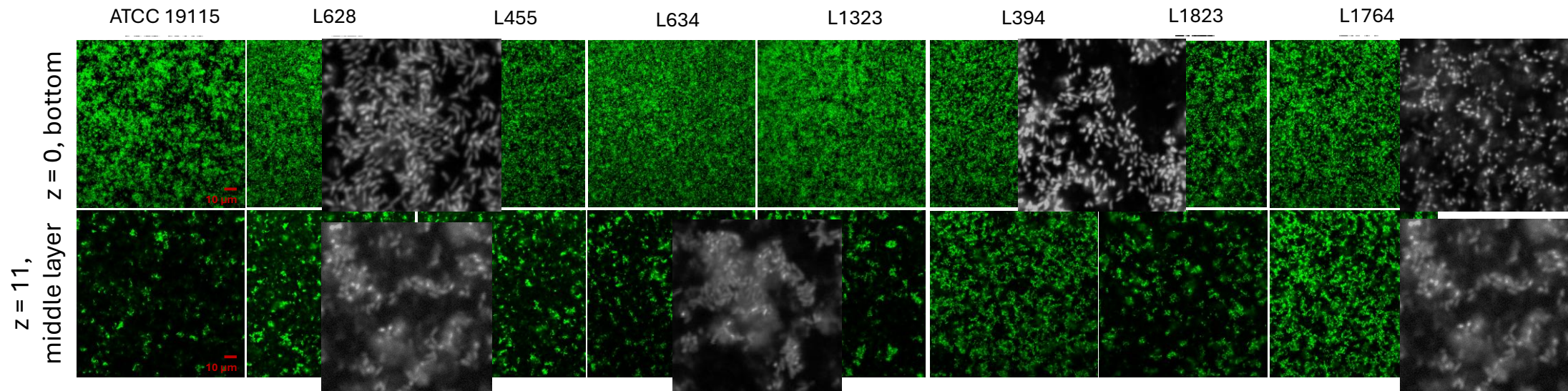


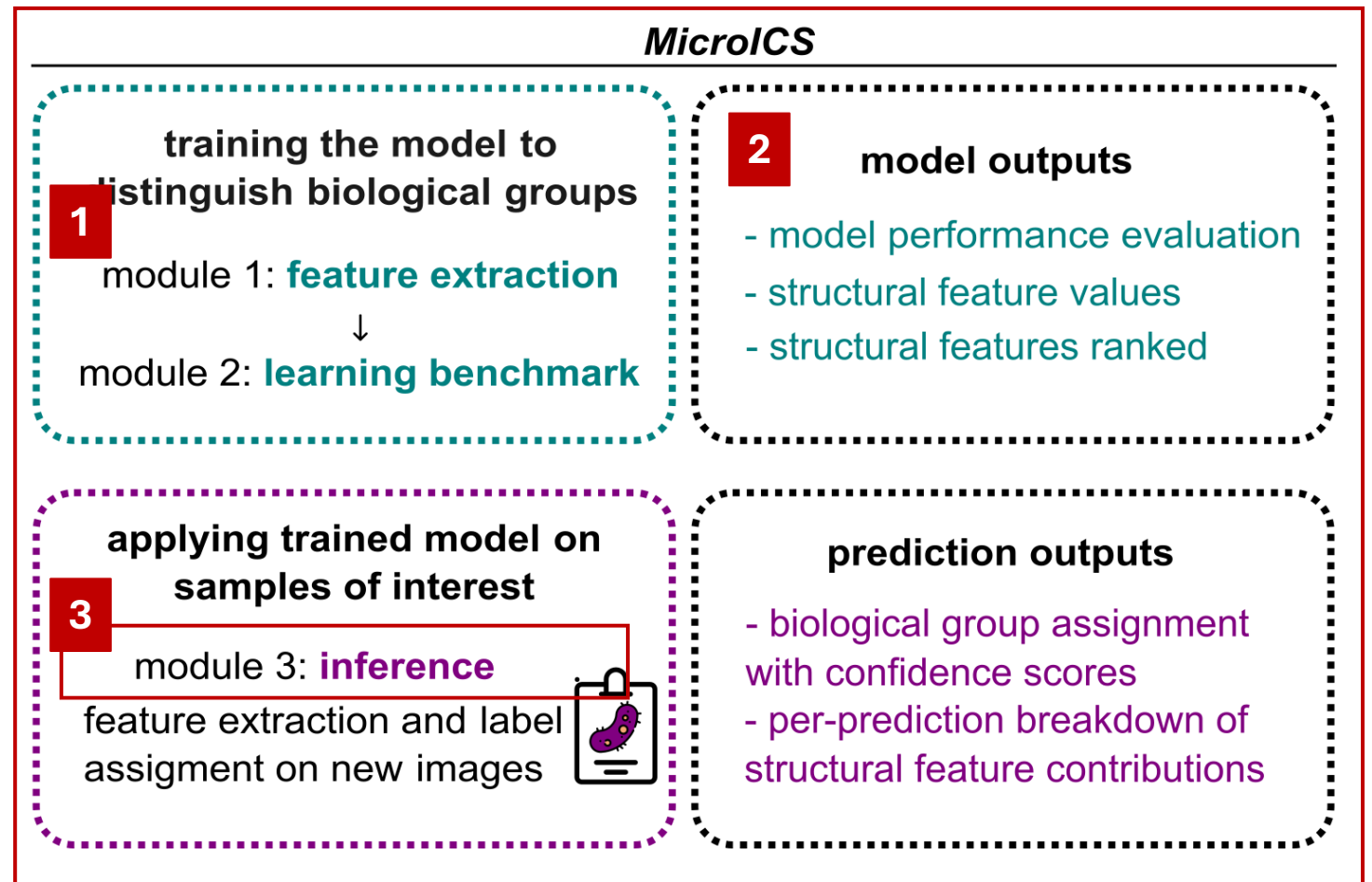
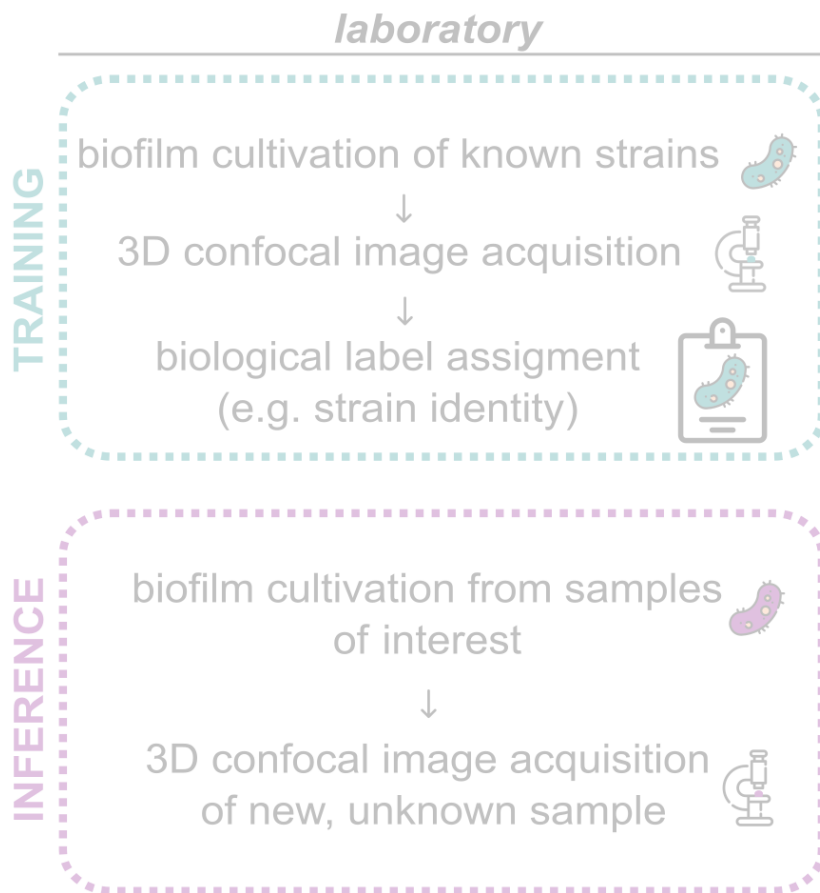
Our first analysis: what the trained model reports.

score	feature name	aggregation type	found among the top-ranking features according to the date	calculated for layer or image
0.08112858	fractal dimension	max	yes	layer
0.06571528	counts (normalized intensity, normalized per layer, threshold < 0.11)	max	no	layer
0.05795771	minimum intensity	std	no	layer
0.03175054	counts (normalized intensity, normalized per layer, threshold < 0.14)	max	no	layer
0.02784039	min prop	min	yes	layer
0.02188584	counts (threshold < 0.06)	median	no	layer
0.02038614	fractal dimension	q90	no	layer
0.01730499	counts (normalized intensity, normalized per layer, threshold < 0.13)	max	no	layer
0.01443651	counts (normalized intensity, normalized per layer, threshold < 0.12)	max	no	layer



Output files: rankings_date.tsv,
Rankings_label.tsv

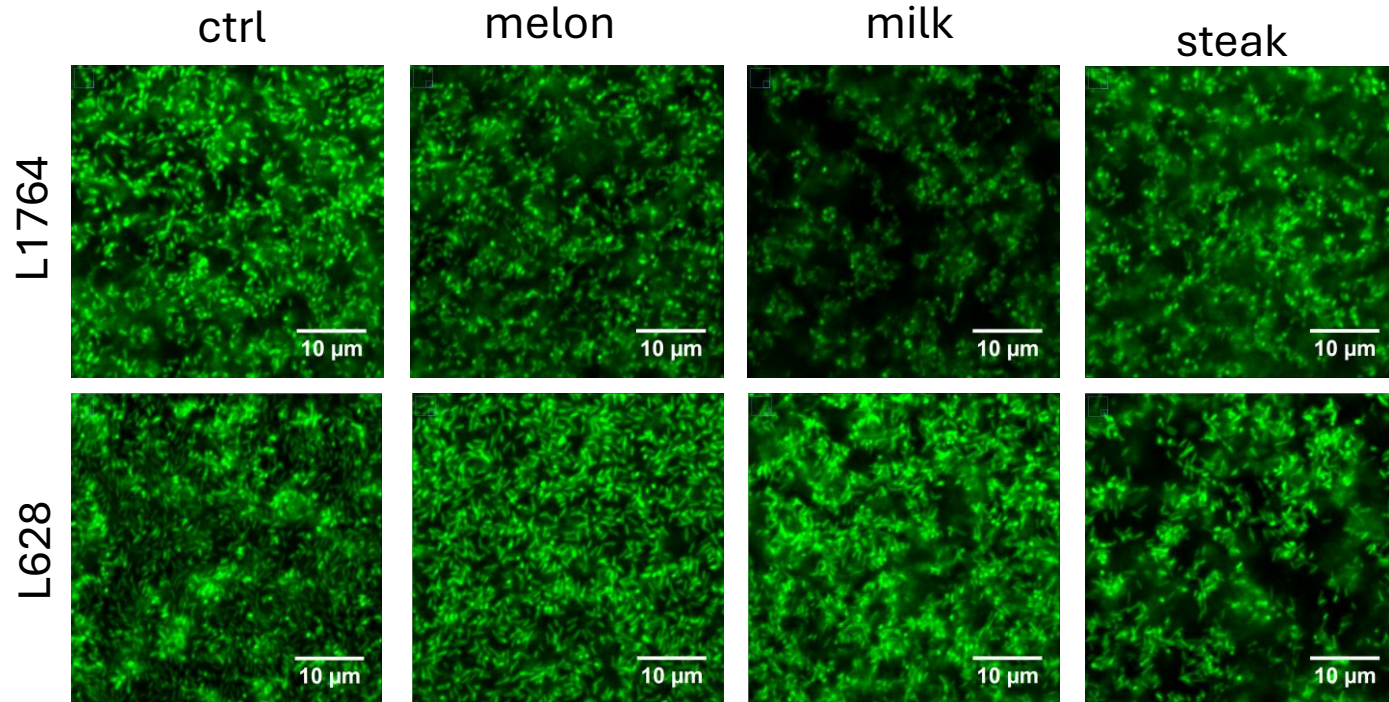




Our first analysis: inference – do food residues affect biofilm structure?

We introduce perturbation by growing biofilms on food residues and want to know how the biofilm structure changes.

Observations from the images from the perturbation experiment:



Our first analysis: inference – do food residues affect biofilm structure?

Generate features for unknown images.

The unknown images are available here:

<https://portal.ijs.si/nextcloud/f/53497958>,

folder

images_generate_features_unlabelled

Download results – you will need feature table for inference!

The image shows a web-based pipeline interface for biofilm analysis. It consists of two main panels. The left panel has four steps: 1. Generate features (with sub-options 'Labelled' and 'Unlabelled', where 'Unlabelled' is selected), 2. Train models, 3. Inference, and 4. All together. The right panel contains configuration options: 'IMAGES TO CLASSIFY' with a 'CHOOSE INFERENCE IMAGES' button; 'PARALLEL WORKERS' set to 4 and 'TOP FEATURES' set to 20; 'CORRELATION |R| THRESHOLD' set to 0.8; and 'Voxel dimensions (µm)' with X, Y, and Z values of 0.13, 0.13, and 0.5 respectively. At the bottom, there is a 'Suggested workers' section, a 'Where do the results go?' section, and a large green 'Run pipeline' button. Numbered callouts 1 through 5 highlight specific elements: 1 points to the 'Unlabelled' option, 2 points to the 'CHOOSE INFERENCE IMAGES' button, 3 points to the 'PARALLEL WORKERS' and 'TOP FEATURES' settings, 4 points to the 'Voxel dimensions' settings, and 5 points to the 'Run pipeline' button.

1 Generate features
Choose whether image filenames contain class labels.

☐ Labelled
For training images

☒ Unlabelled
For inference images

2 **IMAGES TO CLASSIFY** ?
CHOOSE INFERENCE IMAGES
No inference images uploaded.

3 **PARALLEL WORKERS** ? **TOP FEATURES** ?
4 20

4 **CORRELATION |R| THRESHOLD** ?
0.8

5 **Voxel dimensions (µm)** ?
X (µm) Y (µm) Z (µm)
0.13 0.13 0.5

Suggested workers Uses a conservative share of the CPUs available to this container.

Where do the results go?
They appear in [Inspect results](#) on this page. After the run, [Download results ZIP](#) saves a copy to your browser's configured download location; no output folder selection is needed.

Run pipeline
Ready for a new isolated job.

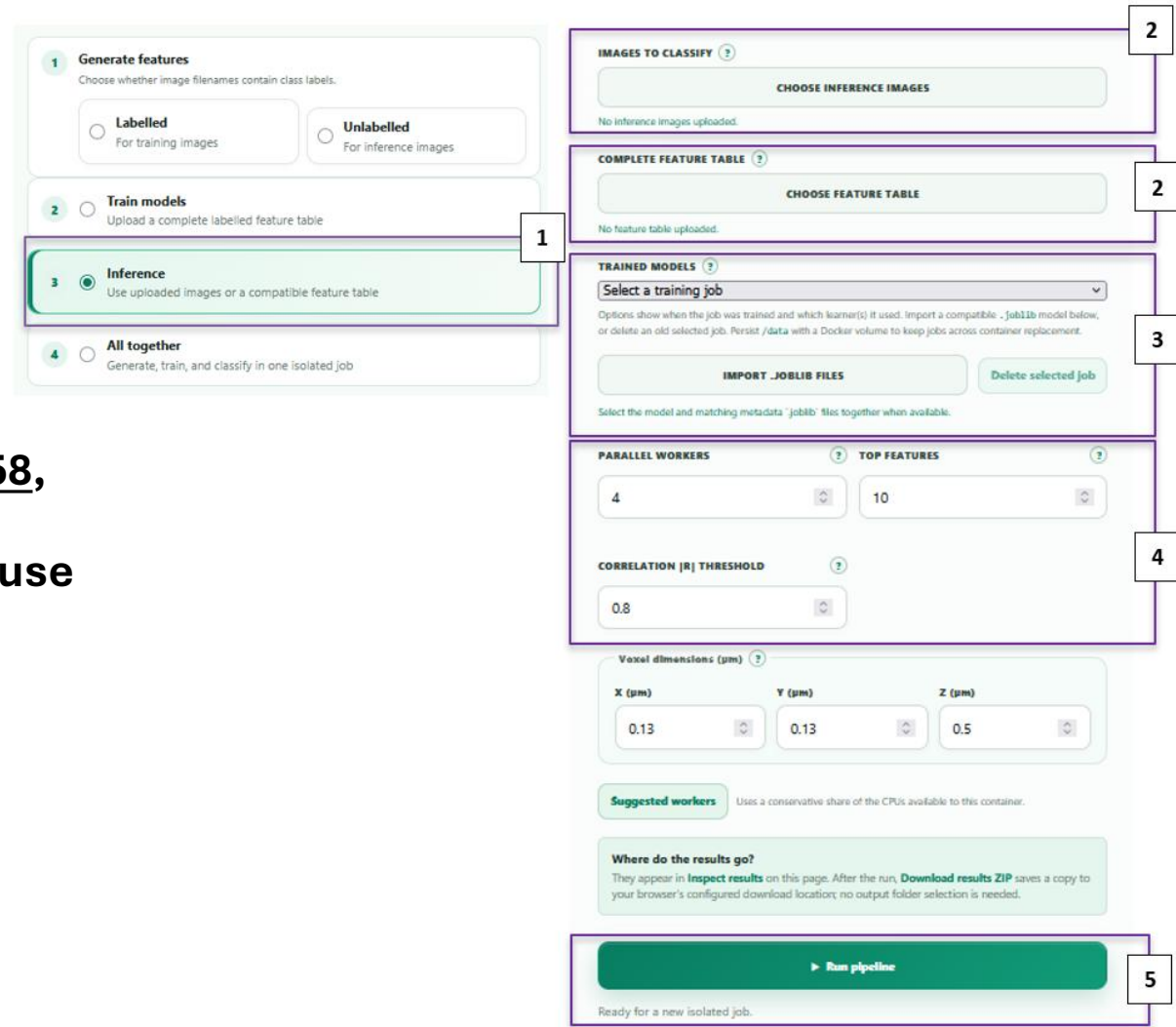
Our first analysis: inference – do food residues affect biofilm structure?

Run inference:

Use prebuild models:

<https://portal.ijis.si/nextcloud/f/53497958>,
folder trained_models_identity ->
models_identity -> mark all of them and use
for inference

Use feature table – results from the
generate features unlabelled



1 **Generate features**
Choose whether image filenames contain class labels.

☐ **Labelled**
For training images

☐ **Unlabelled**
For inference images

2 **Train models**
Upload a complete labelled feature table

3 **Inference**
Use uploaded images or a compatible feature table

4 **All together**
Generate, train, and classify in one isolated job

2 **IMAGES TO CLASSIFY**
CHOOSE INFERENCE IMAGES
No inference images uploaded.

2 **COMPLETE FEATURE TABLE**
CHOOSE FEATURE TABLE
No feature table uploaded.

3 **TRAINED MODELS**
Select a training job
Options show when the job was trained and which learner(s) it used. Import a compatible .joblib model below, or delete an old selected job. Persist /data with a Docker volume to keep jobs across container replacement.
IMPORT .JOBLIB FILES Delete selected job
Select the model and matching metadata .joblib files together when available.

4 **PARALLEL WORKERS** **TOP FEATURES**
4 10

4 **CORRELATION [R] THRESHOLD**
0.8

Voxel dimensions (µm)
X (µm) Y (µm) Z (µm)
0.13 0.13 0.5

Suggested workers Uses a conservative share of the CPUs available to this container.

Where do the results go?
They appear in **Inspect results** on this page. After the run, **Download results ZIP** saves a copy to your browser's configured download location; no output folder selection is needed.

5 **Run pipeline**
Ready for a new isolated job.

Our first analysis: inference – do food residues affect biofilm structure?

Download inference results

Look for file “predictions.tsv”, open it by dragging it into excel file

On the right is the summary of the big run (350 images): cca 50 repetitions per condition + controls

Check controls!

Check predicted classes!

strain	treatment	voted class							
		L1764	L455	L1323	L394	L628	ATCC 19115	L1823	L634
L1764	control	19	2	1	3	0	0	0	0
L1764	melon extract coating	46	0	0	4	0	0	0	0
L1764	milk coating	46	0	0	2	0	0	2	0
L1764	steak tartare extract coating	50	0	0	0	0	0	0	0
L628	control	0	0	0	0	24	0	1	0
L628	melon extract coating	21	14	0	7	5	0	2	0
L628	milk coating	42	0	0	4	0	0	4	0
L628	steak tartare extract coating	47	0	0	1	0	0	2	0

Our first analysis: inference – do food residues affect biofilm structure?

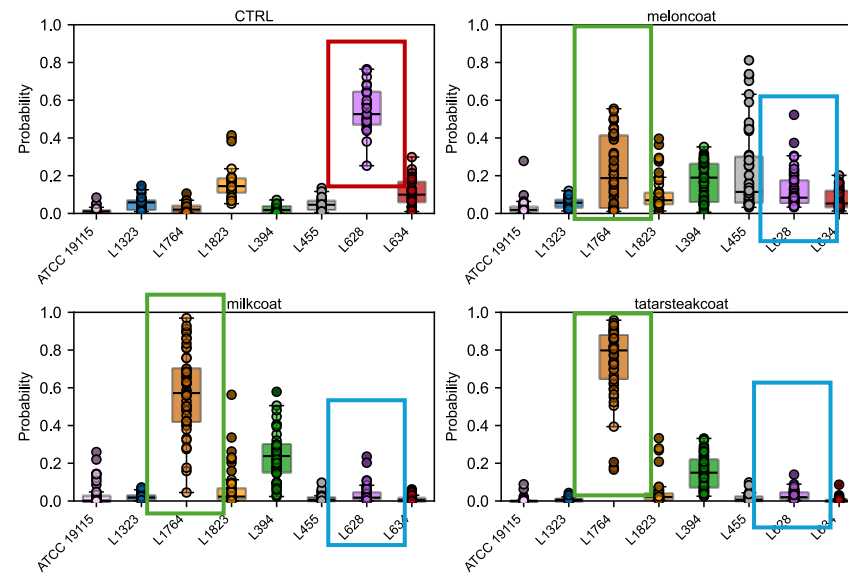
File: “probabilities.tsv”

Check controls (red).

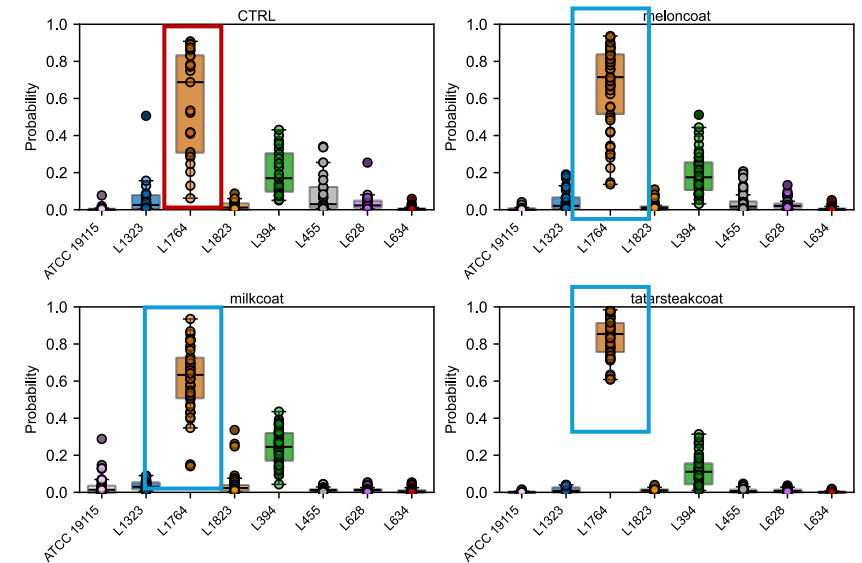
Check, by identity, the class to which the strain belongs (blue).

Check the predicted class (green).

L628



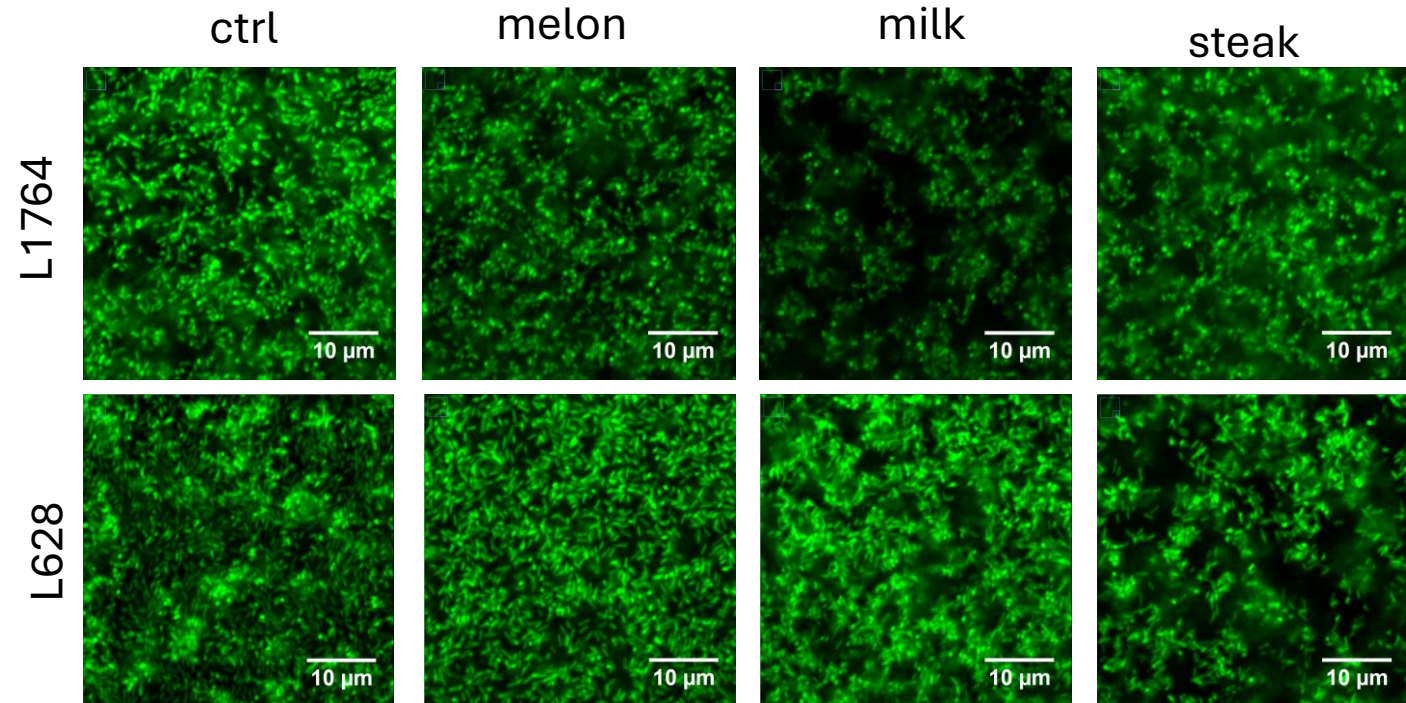
L1764



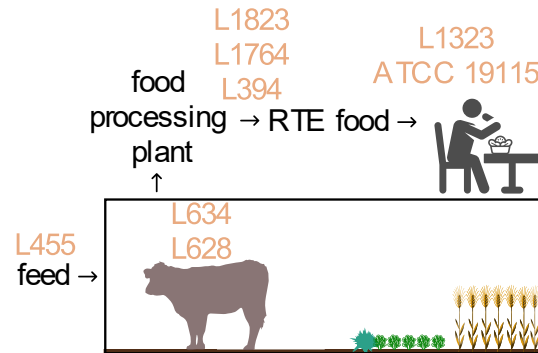
Our first analysis: inference – do food residues affect biofilm structure?

“Not plastic”: Stays most similar to itself.

“Plastic”: melon shifts structure towards L1764, but in fact differs from all strains in the dataset.
Milk and steak: L628 is becoming similar to strain L1764 (smaller clusters, no confluent growth).
Clear shift in structure!

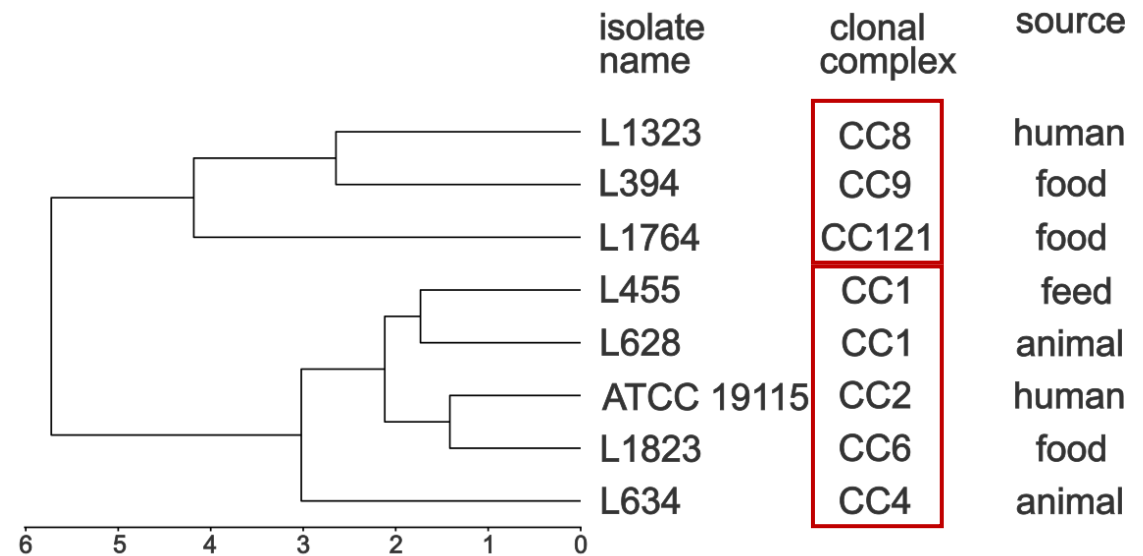


Our second analysis: use the same dataset for different question



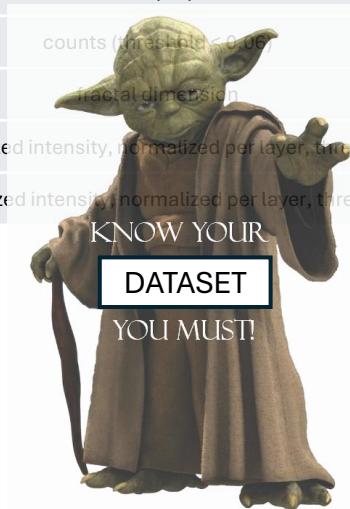
Can we group the strains by CC according to whether they are clinically relevant or not, in order to link biofilm structure to clinical relevance?

- Assigning label:
 - Do you remember what you change in the name of the image?
 - Is clinical association the ground truth?
- Number of classes
- Number of strains per class



Our second analysis: use the same dataset for different question – shift in top features

IDENTITY	feature name	aggregation type	found among the top-ranking features according to the date	calculated for layer or image
	fractal dimension	max	yes	layer
	counts (normalized intensity, normalized per layer, threshold < 0.11)	max	no	layer
	minimum intensity	std	no	layer
	counts (normalized intensity, normalized per layer, threshold < 0.14)	max	no	layer
	min prop			
	counts (normalized intensity, normalized per layer, threshold < 0.06)			
	fractal dimension			
	counts (normalized intensity, normalized per layer, threshold < 0.13)			
	counts (normalized intensity, normalized per layer, threshold < 0.12)			



CLINICAL RELEVANCE	feature name	aggregation type	found also in ranking according to the date among 81 features	calculated for layer or image
	counts (normalized intensity, normalized per layer, threshold < 0.19)	max	no	layer
	counts (normalized intensity, normalized per layer, threshold < 0.11)	median	no	layer
	counts (normalized intensity, normalized per layer, threshold < 0.99)	median	no	layer
	counts (threshold < 0.99)	q90	yes	layer
	counts (normalized intensity, normalized per layer, threshold < 0.29)	max	no	layer
	counts (threshold < 0.60)	std	no	layer
	counts (normalized intensity, normalized per layer, threshold < 0.20)	max	no	layer
	counts (normalized intensity, normalized per layer, threshold < 0.03)	max	no	layer
	counts (threshold < 0.06)	var	no	layer
	counts (normalized intensity, normalized per layer, threshold < 0.12)	median	no	layer

Abundance related features

Is this a legitimate conclusion in terms of machine learning?

Is this a legitimate conclusion from a biological perspective (hint: strains per class, label type)?

Wrap up



Training school

The purple path is the worked *Listeria* example we run together.

Think about how it applies to your own work

The green path is your own work, carried from the start so that the two meet at the end.



DECIDE & FRAME · Obj 1–2

What to measure?

*defining your own question
defining your own label*



DESIGN & ACQUIRE · Obj 3

How to measure?

*planning your experiments
and imaging settings*



RUN MicroICS · Obj 4

How to analyse?

*independently running the
software*



READ & INTERPRET · Obj 5–6

How to interpret results?

*interpreting your own output
how can the output help me
to understand my dataset*



KNOW THE LIMITS · Obj 7

What are the limitations of the pipeline?

*identifying the follow-up
experiments*

The destination

You can see how to apply MicroICS to your own organism, treatment, and biological question to help you generate hypotheses for further work.